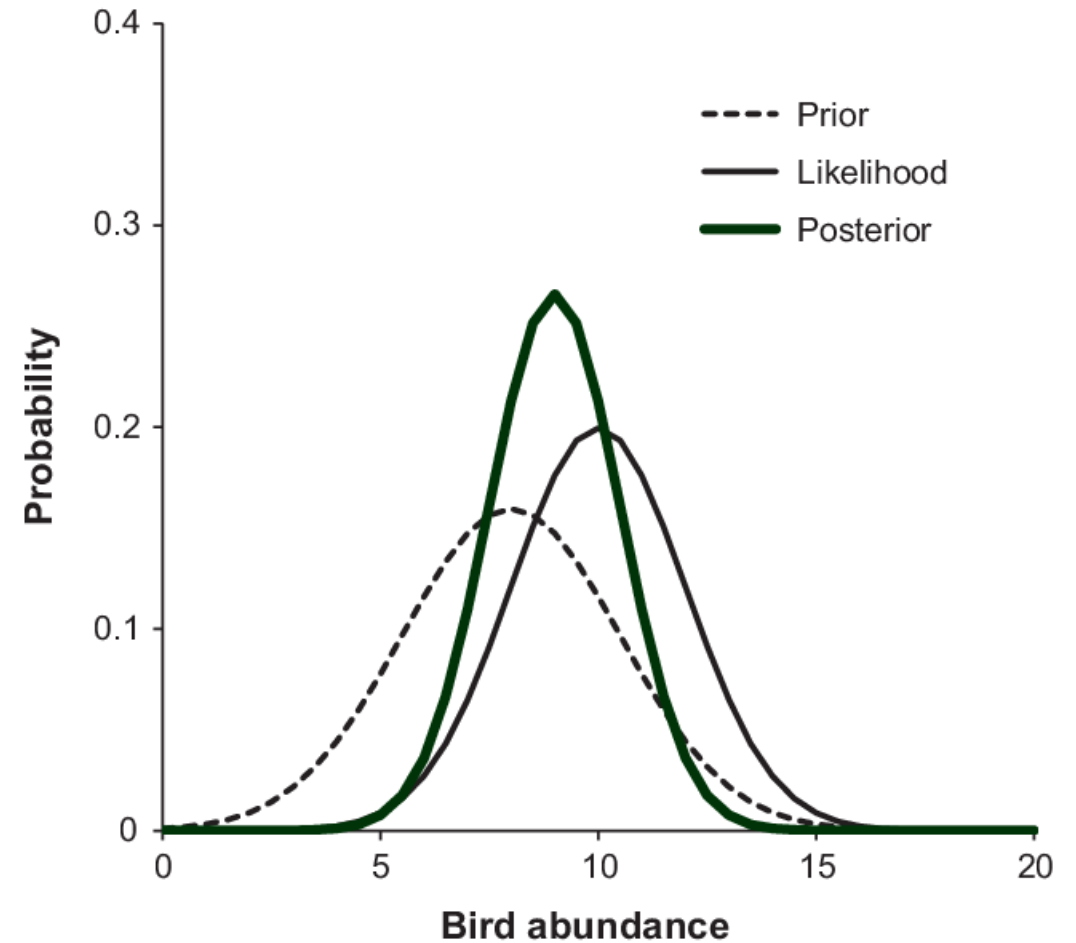
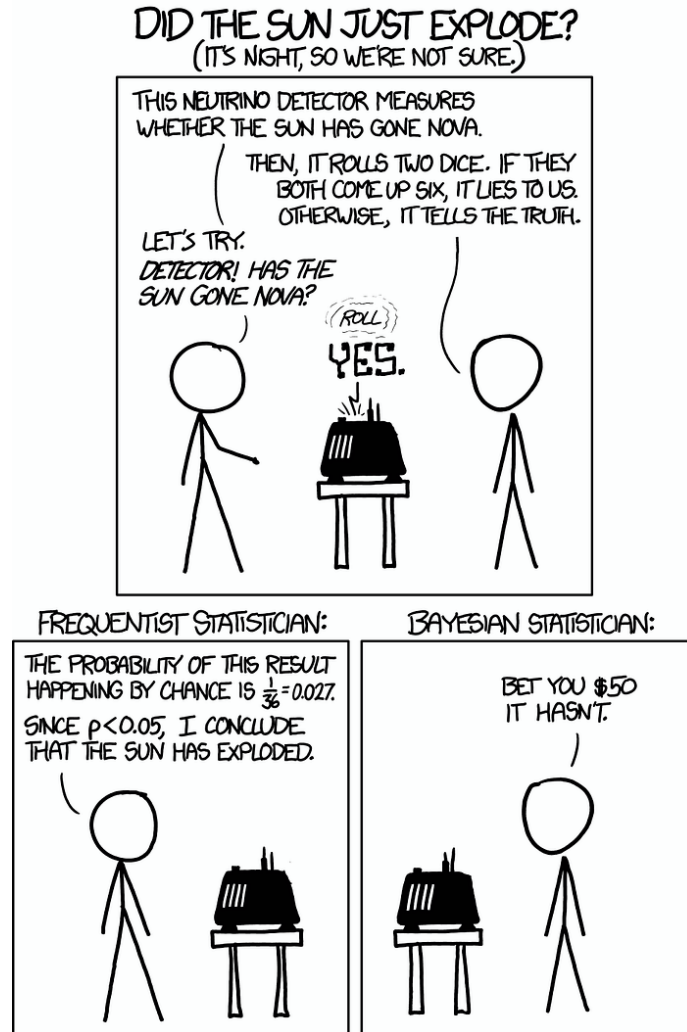


Bayesian statistics and priors



No class next Thursday

Office hours (Stone 307A):

MF 0930-1100

W 1100-1300

We're going to talk about a LOT of stuff today!

We won't master it all

The key idea is to BEGIN to think about how priors and our data interact to inform our final estimate in a Bayesian analysis

These are concepts we'll revisit over and over again through the semester

Outline

1. BAYES THEOREM

- *The technical backbone*

2. STARTING SIMPLE

- *What is a prior? silly examples!*
- *A back of the envelope wildlife example... beta priors*

3. ADDING COMPLEXITY (also in future weeks)

- *Key criticisms, and why they lead to problems*
- *Unexpectedly bad priors (Northrup & Gerber, 2018) and philosophy*

BAYES THEOREM

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

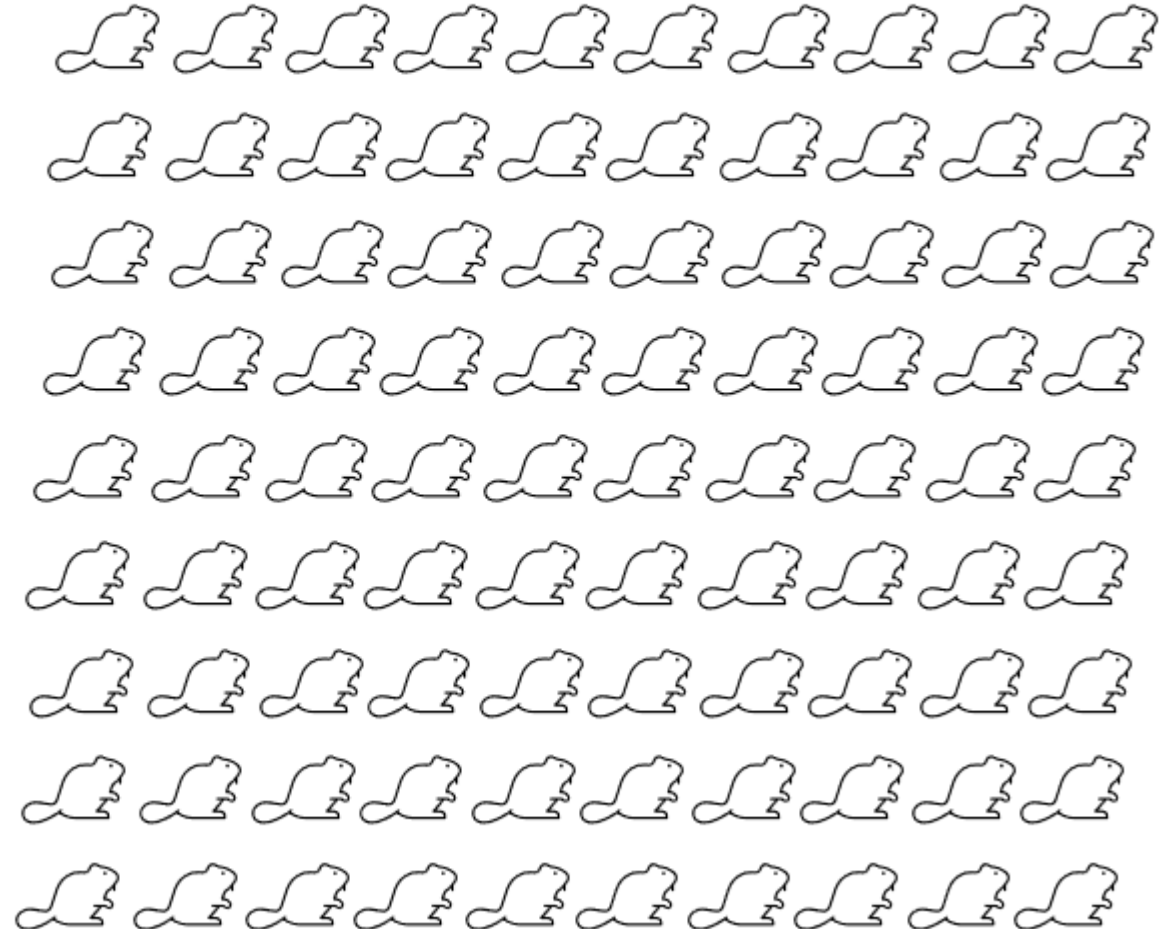
BAYES THEOREM: a quick example from disease ecology

Actually sick

False positives



- Imagine we test 100 beavers for a disease.
- If the disease is present, we observe it (no false negatives!)
- There is an 9% false positive rate...



BAYES THEOREM: a quick example from disease ecology

- Imagine we test 100 beavers for a disease. 1% of beavers are sick.

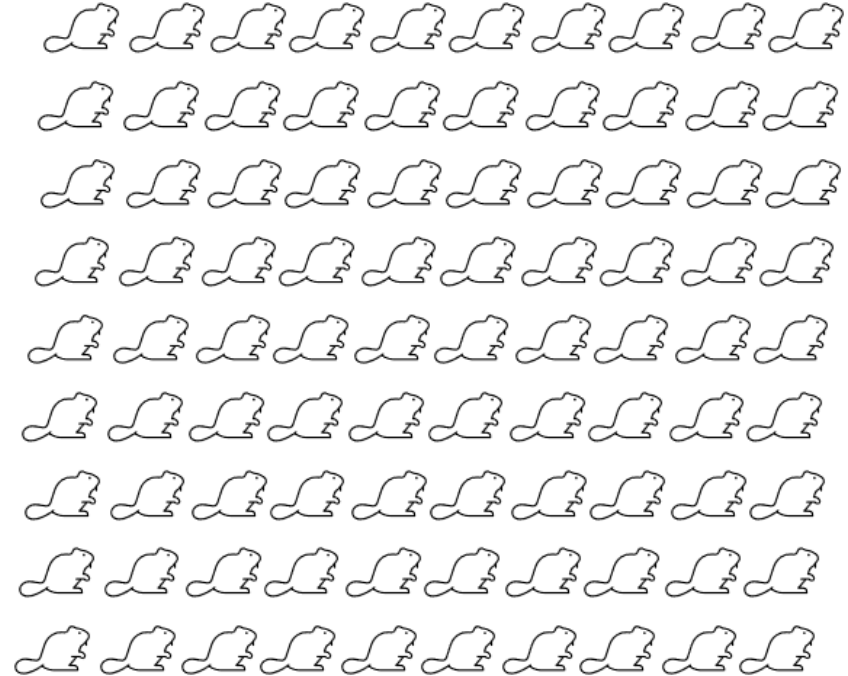
Actually sick



False positives



- If the disease is present, we observe it (no false negatives!)
- There is an ~9% false positive rate...



$$P(Disease|+) = \frac{P(+|Disease) \times P(Disease)}{P(+)}$$

BAYES THEOREM: a quick example from disease ecology

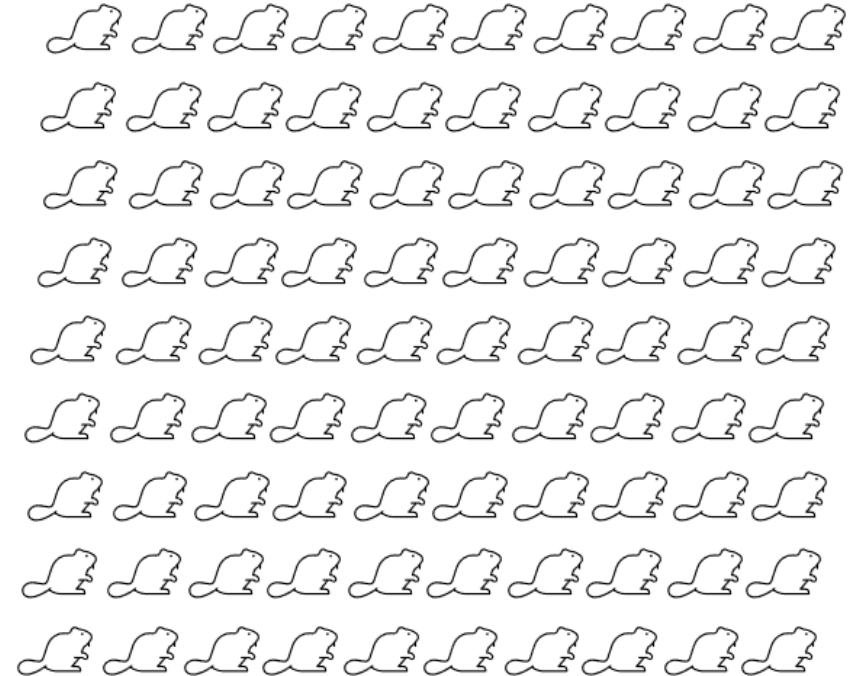
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Actually sick



False positives



$$P(D|+) \approx 0.1$$

BAYES THEOREM: a quick example from disease ecology

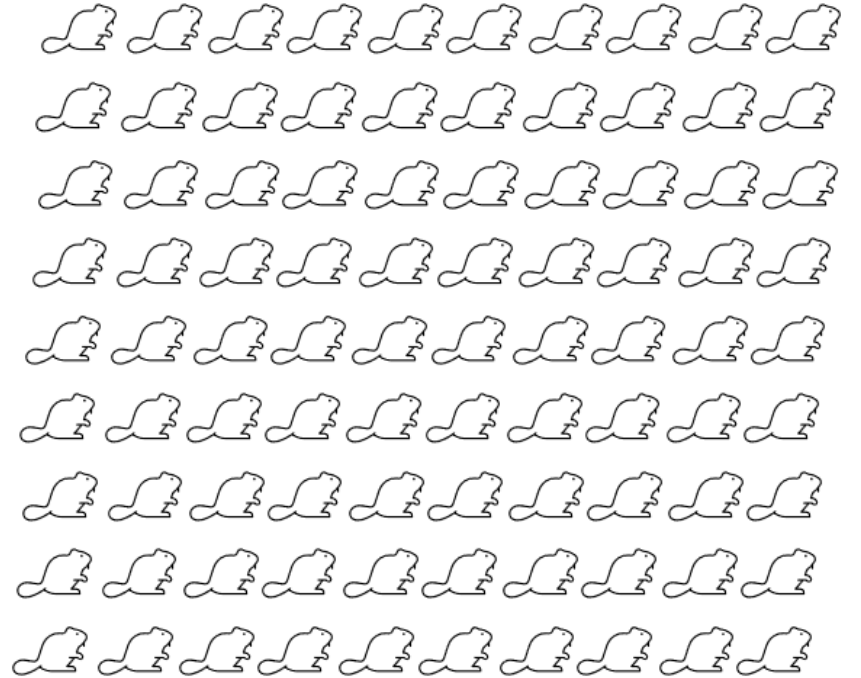
- Imagine we test 100 beavers for a disease. **1% of beavers are sick.**
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- There is an ~9% false positive rate...

$$P(D|+) = \frac{P(+|D) \times P(D)}{P(+)}$$
$$P(D|+) = \frac{1 \times 0.01}{(1 * 0.01) + (0.09 * .99)}$$

Actually sick



False positives



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BAYES THEOREM: a quick example from disease ecology

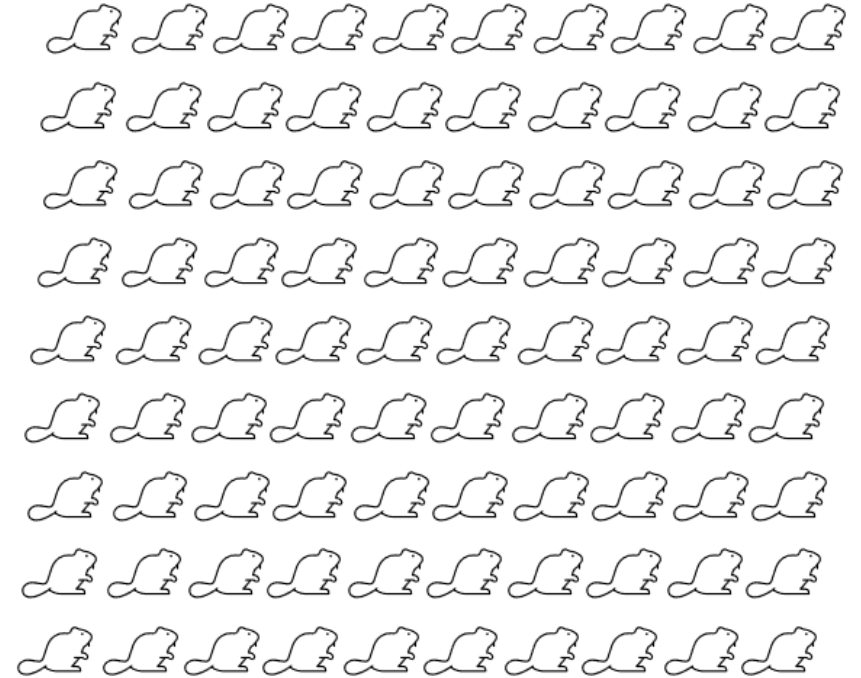
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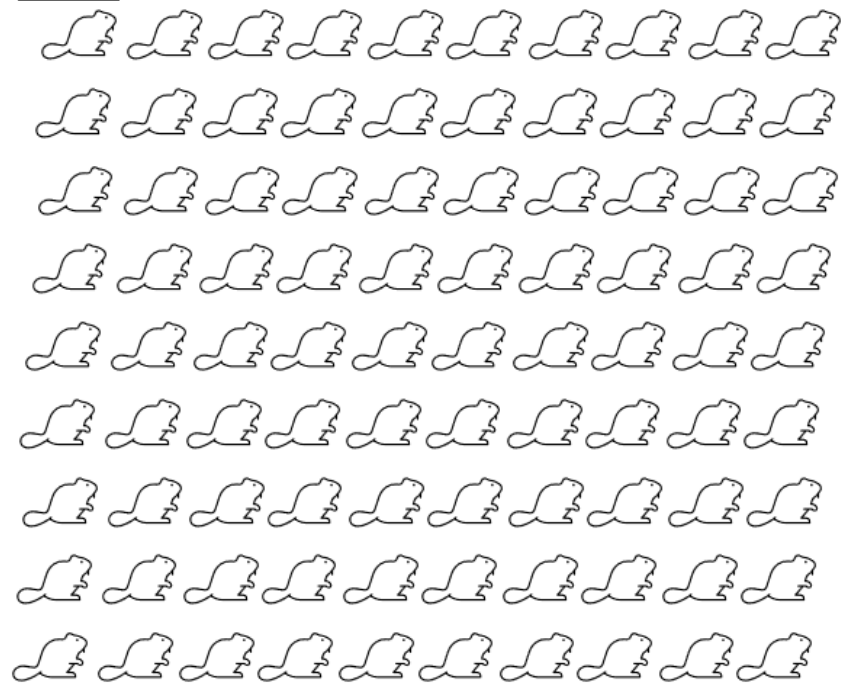
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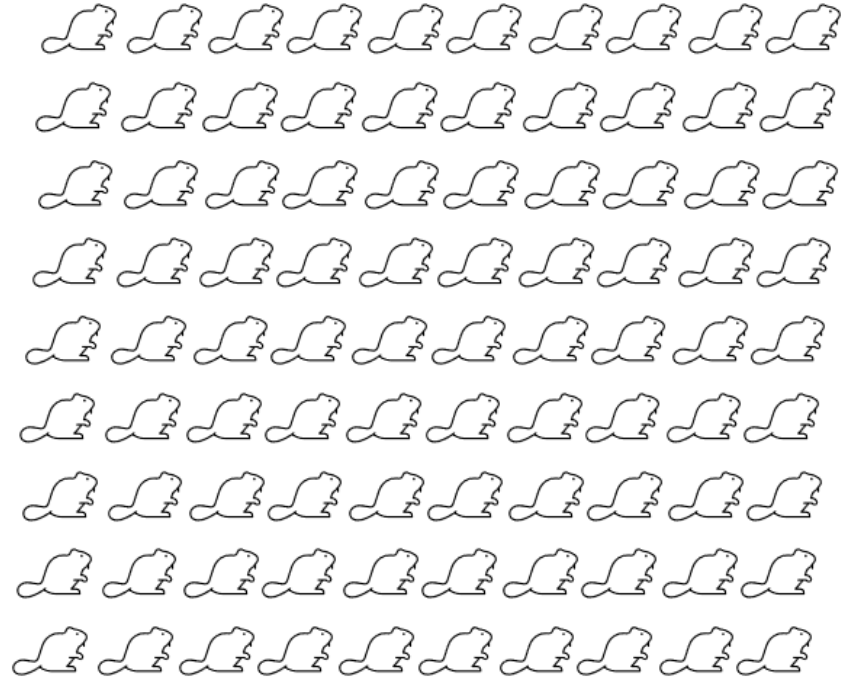
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Actually sick



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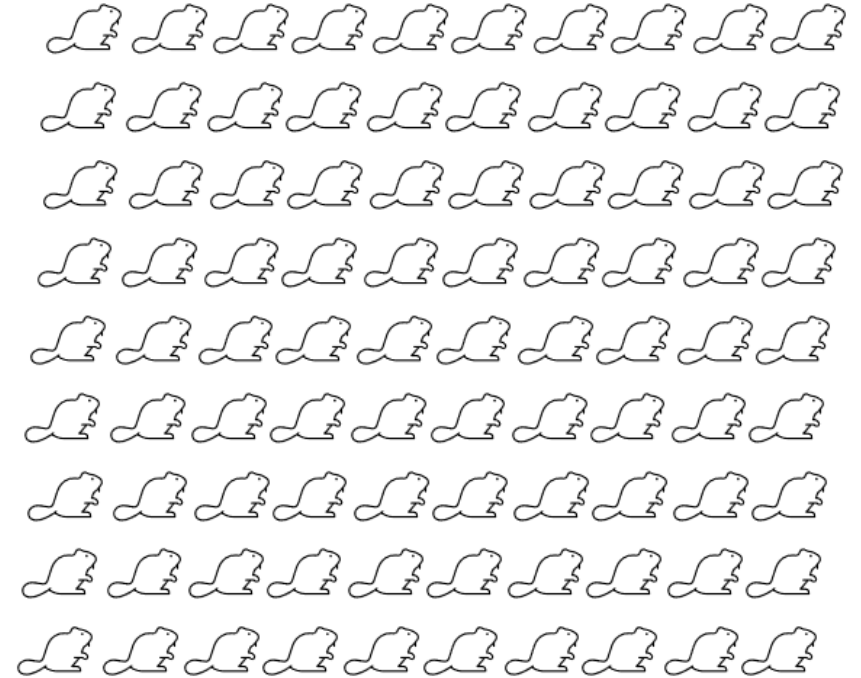
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Actually sick



False positives



$$P(D|+) \approx 0.1$$

BAYES THEOREM

$$P(\boldsymbol{\theta}|y) = \frac{P(y|\boldsymbol{\theta}) \times P(\boldsymbol{\theta})}{P(y)}$$

BAYES THEOREM

$$P(\theta|y) = \frac{P(y|\theta) \times P(\theta)}{P(y)}$$

θ : parameter(s) of interest

y : data

BAYES THEOREM

Likelihood [these are the estimates you'd get from `lm()`]

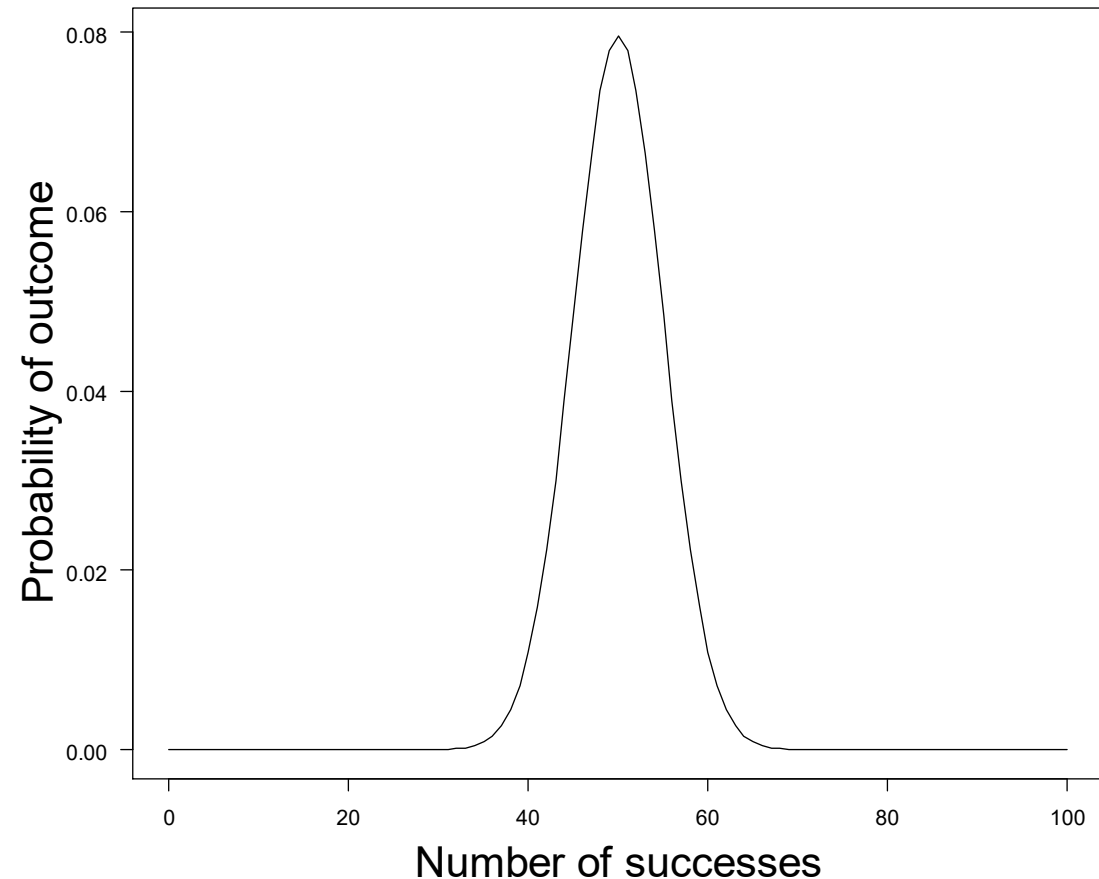
$$P(\theta|y) = \frac{P(y|\theta) \times P(\theta)}{P(y)}$$

θ : parameter(s) of interest

y : data

What is a **likelihood**?

- Probability for getting a result for a given value of parameter(s)



Flipping a coin 100 times with 50:50 chance of heads

BAYES THEOREM

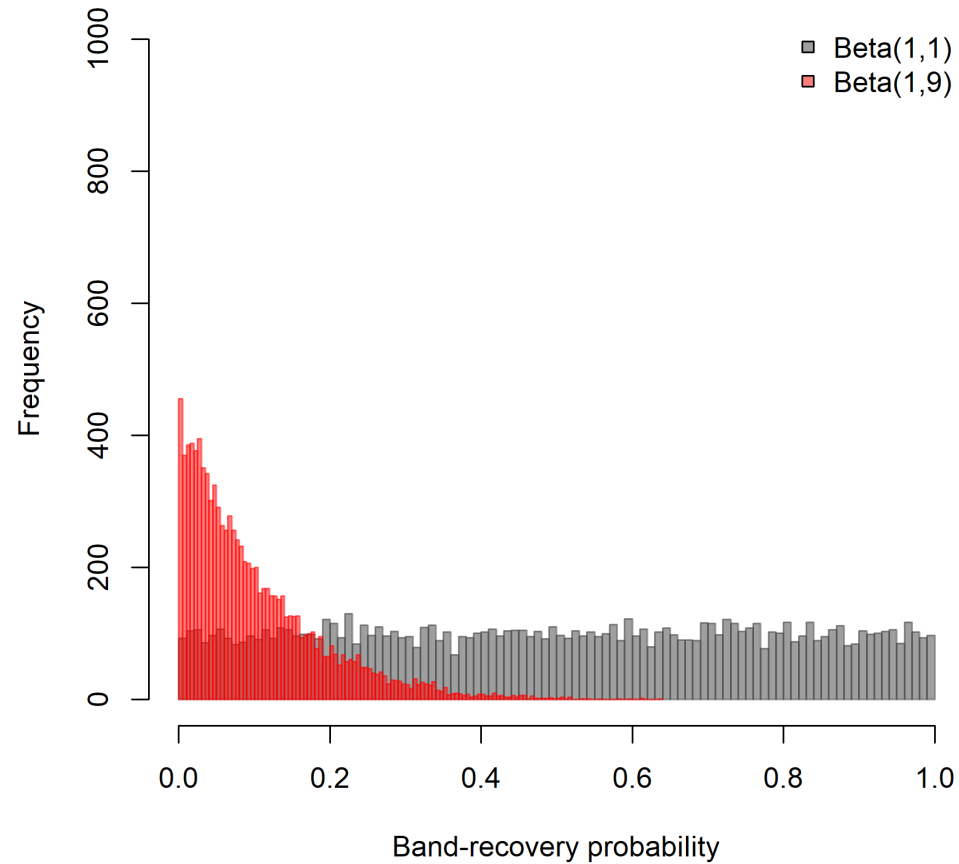
$$P(\theta|y) = \frac{\overset{\text{Likelihood}}{\boxed{P(y|\theta)}} \times \overset{\text{Prior}}{\boxed{P(\theta)}}}{P(y)}$$

θ : parameter of interest

y : data

What is a prior?

- Your prior belief in the distribution of a parameter.



BAYES THEOREM

$$P(\theta|y) = \frac{\overset{\text{Likelihood}}{\boxed{P(y|\theta)}} \times \overset{\text{Prior}}{\boxed{P(\theta)}}}{\underset{\text{Marginal}}{\boxed{P(y)}}}$$

θ : parameter of interest

y : data

What is **marginal**?

- The marginal probability of the data!
- i.e., the sum of the probability of the different ways you could observe the data

BAYES THEOREM

$$\text{Posterior} \quad P(\theta|y) = \frac{\text{Likelihood} \quad P(y|\theta) \times \text{Prior} \quad P(\theta)}{\text{Marginal} \quad P(y)}$$

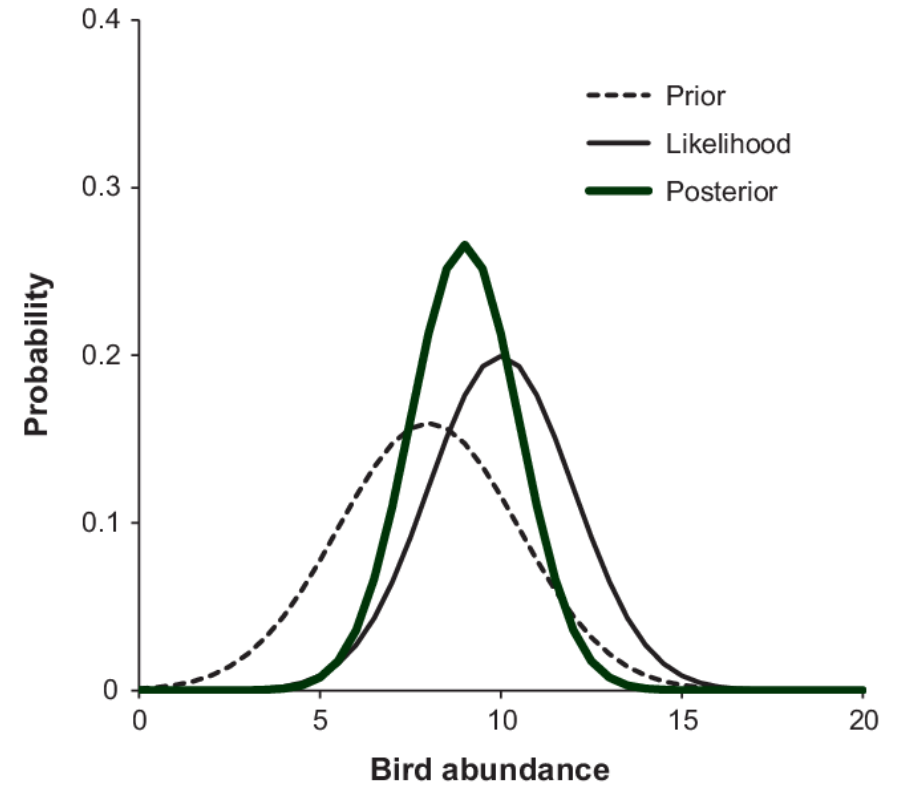
θ : parameter of interest

y : data

What is a **posterior**?

- The distribution of a parameter given your prior belief and the data.

Our estimate



BAYES THEOREM: let's work through some fun, simple e.g.'s



The case of the credulous crackpot

$$P(\theta|y) = \frac{P(y|\theta) \times P(\theta)}{P(y)}$$

BAYES THEOREM: let's work through a silly, intuitive example



$$P(\text{bigfoot}|+) = \frac{P(+|\text{bigfoot}) \times P(\text{bigfoot})}{P(+)}$$

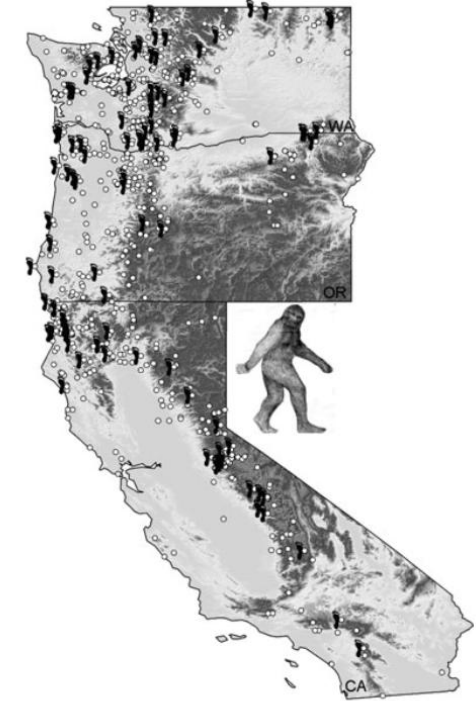


Figure 1 Map of Bigfoot encounters from Washington, Oregon and California used in the analyses. Points represent visual/auditory detection, and foot symbols represent coordinates where footprint data were available. Shading indicates topography, with lighter values representing lower elevations.

Lozier **(2009)** *Journal of Biogeography*

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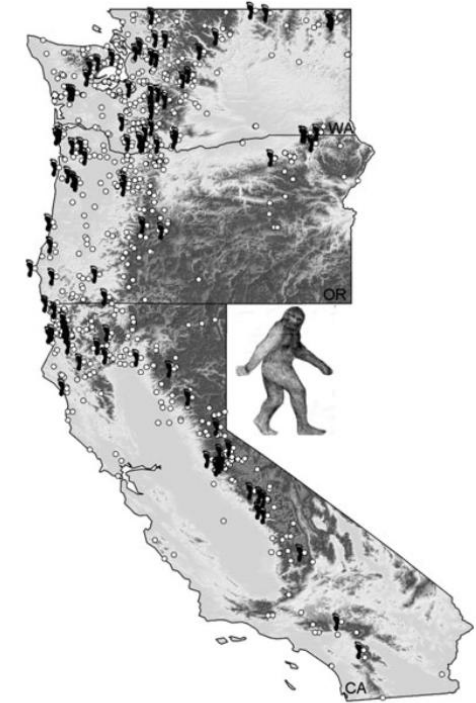


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Lozier (**2009**) *Journal of Biogeography*

$$P(\theta|y) = \frac{P(y|\theta) \times P(\theta)}{P(y)}$$

BAYES THEOREM: let's work through a silly, intuitive example



Prior

$$P(\text{bigfoot}|+) = \frac{P(+|\text{bigfoot}) \times P(\text{bigfoot})}{P(+)}$$

Prior: the probability of an event pre-data

Lozier (2009) *Journal of Biogeography*

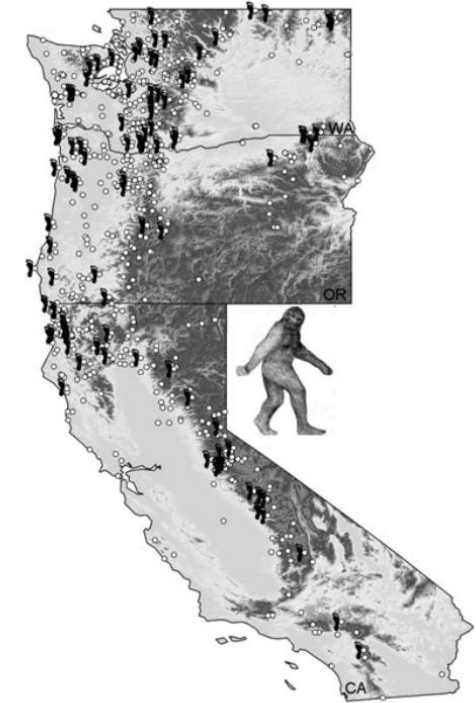


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$$P(\theta|y) = \frac{P(y|\theta) \times P(\theta)}{P(y)}$$

BAYES THEOREM: let's work through a silly, intuitive example



$$P(\text{bigfoot}|+) = \mathbf{0} = \frac{P(+|\text{bigfoot}) \times \mathbf{0}}{P(+)}$$

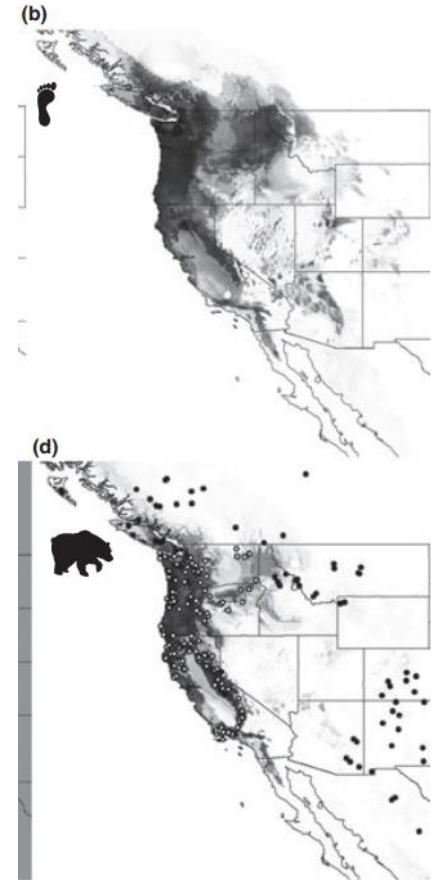
Bigfoot does not exist, zero divided by whatever = 0... analysis complete

$$P(\theta|y) = \frac{P(y|\theta) \times P(\theta)}{P(y)}$$

BAYES THEOREM: what about the **marginal**?



$$P(\text{bigfoot}|+) = \frac{P(+|\text{bigfoot}) \times P(\text{bigfoot})}{P(+)}$$



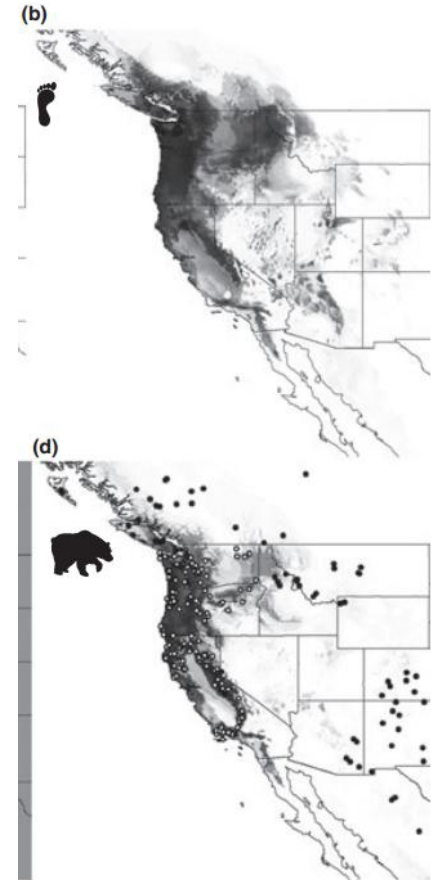
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BAYES THEOREM: what about the **marginal**?



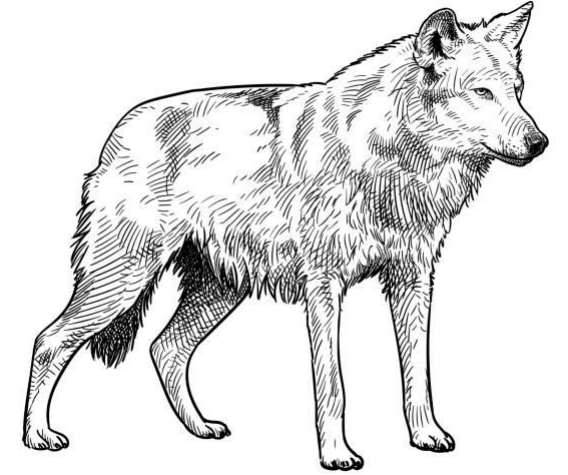
$$P(\text{bigfoot}|+) = \frac{P(+|\text{bigfoot}) \times P(\text{bigfoot})}{P(+ \cap \text{bigfoot}) + P(+ \cap \text{bear})}$$



Lozier (2009) *Journal of Biogeography*

$$P(\theta|y) = \frac{P(y|\theta) \times P(\theta)}{P(y)}$$

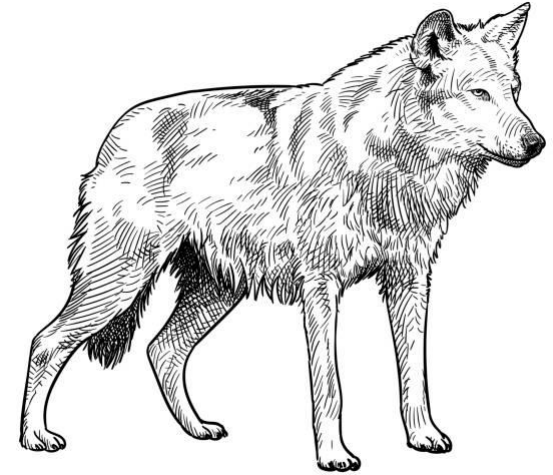
The credulous crackpot strikes again!



$$P(\theta|y) = \frac{P(y|\theta) \times P(\theta)}{P(y)}$$

BAYES THEOREM: let's work through a better example

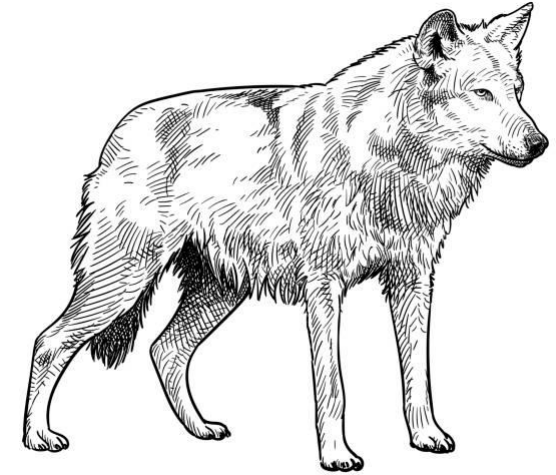
$$P(\text{wolf}|+) = \frac{P(+|\text{wolf}) \times P(\text{wolf})}{P(+)}$$



Lozier (**2009**) *Journal of Biogeography*

BAYES THEOREM: let's work through a better example

$$P(\text{wolf}|+) = \frac{P(+|\text{wolf}) \times \overset{\text{Prior}}{P(\text{wolf})}}{\underset{\text{Marginal}}{P(+)}}$$

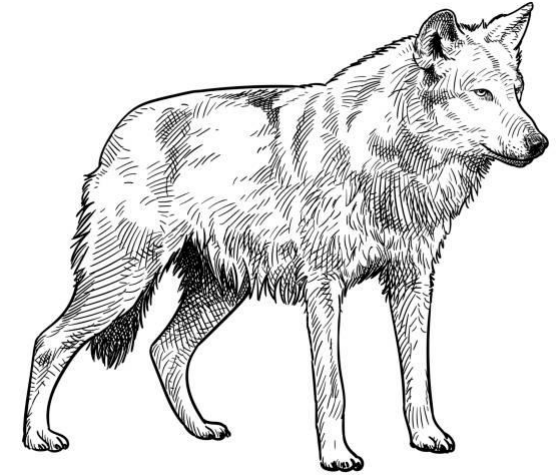


BAYES THEOREM: let's work through a better example

$$P(\text{wolf}|+) = \mathbf{0} = \frac{P(+|\text{wolf}) \times \overset{\text{Prior}}{\boxed{P(\text{wolf})}}}{\underset{\text{Marginal}}{\boxed{P(+)}}}$$

Prior: probability of the event

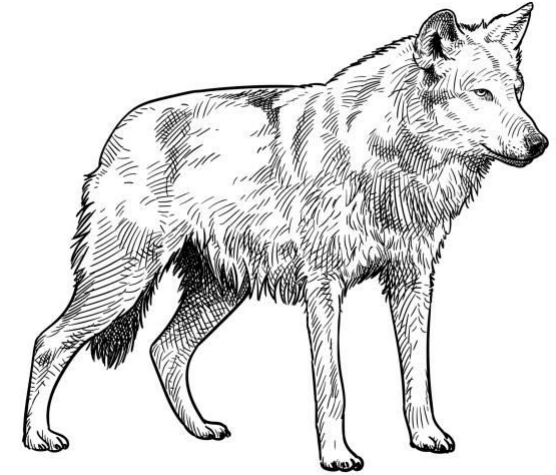
Marginal: the probability of the data...



BAYES THEOREM: let's make some assumptions

$$P(\text{wolf}|+) = \mathbf{0} = \frac{P(+|\text{wolf}) \times \text{Prior}}{\text{Marginal}}$$

The equation shows the calculation of the posterior probability $P(\text{wolf}|+)$. The numerator is the product of the likelihood $P(+|\text{wolf})$ and the prior $P(\text{wolf})$. The denominator is the marginal probability $P(+ \cap \text{wolf}) + P(+ \cap \text{coyote})$. The prior $P(\text{wolf})$ is highlighted with a blue dashed box, and the denominator is highlighted with an orange dashed box.

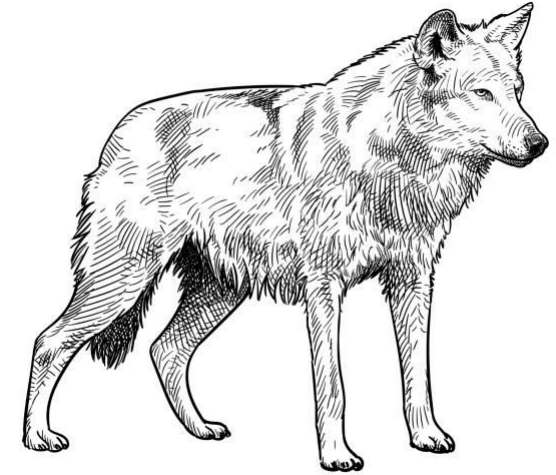


We'll assume they actually saw a large wild canid (i.e., either coyote or wolf)

BAYES THEOREM: let's make some assumptions

$$P(\text{wolf}|+) = \mathbf{0} = \frac{P(+|\text{wolf}) \times \text{Prior}}{\text{Marginal}}$$

The equation shows the posterior probability of a wolf given a positive test result. The numerator is the product of the likelihood $P(+|\text{wolf})$ and the prior probability $P(\text{wolf})$, which is labeled "Prior" in blue. The denominator is the sum of the joint probabilities for wolf and coyote, $P(+ \cap \text{wolf}) + P(+ \cap \text{coyote})$, which is labeled "Marginal" in orange. The result is 0, indicating that the prior probability of a wolf is 0.



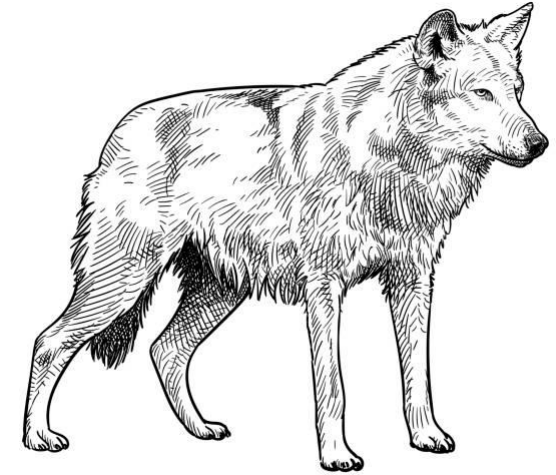
Prior: let's imagine a 2:1 coyote:wolf ratio

Marginal: if wolf, wolf, if coyote, 50:50

BAYES THEOREM: let's make some assumptions

$$P(\text{wolf}|+) = \mathbf{0.5} = \frac{P(+|\text{wolf}) \times \text{Prior}}{\text{Marginal}}$$

The equation shows the calculation of the posterior probability $P(\text{wolf}|+)$ as 0.5. The numerator is the product of the likelihood $P(+|\text{wolf})$ and the prior probability, which is 0.33. The denominator is the marginal probability, which is the sum of 0.33 and 0.33.



Prior: let's imagine a 2:1 coyote:wolf ratio

Marginal: if wolf -> wolf, if coyote, 50:50

BAYES THEOREM: let's make some assumptions

$$P(\text{wolf}|+) = \mathbf{0.5} = \frac{P(+|\text{wolf}) \times \mathbf{0.33}}{\mathbf{0.33 + 0.33}}$$

Prior

Marginal

Prior: let's imagine a 2:1 coyote:wolf ratio

Marginal: if wolf -> wolf, if coyote, 50:50



Wolf

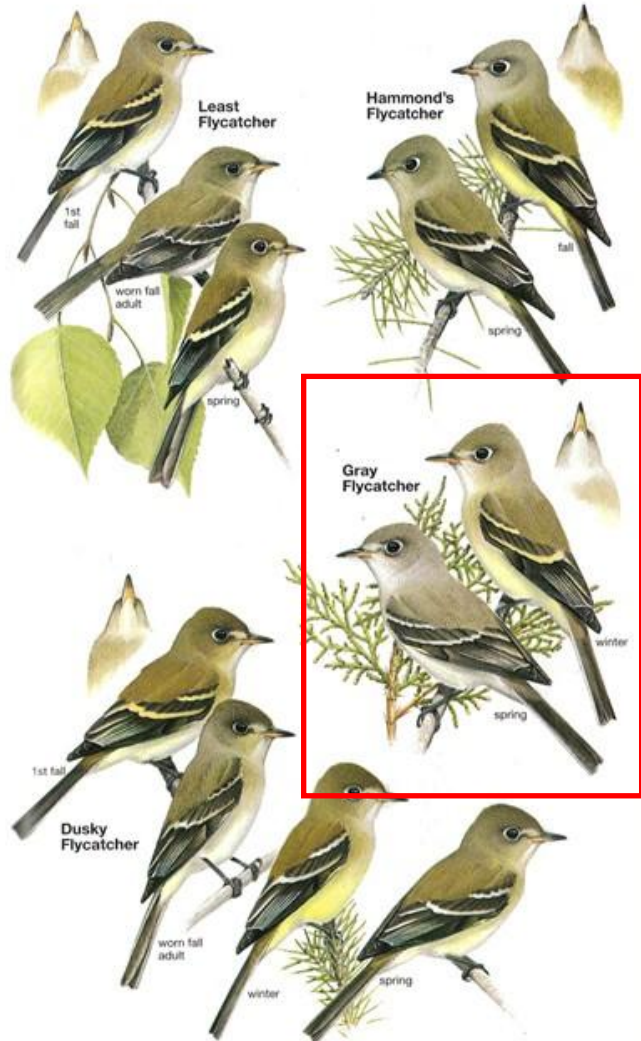


Misidentified as
wolf (coyote)



Coyote

BAYES THEOREM: a complex/realistic example (*Empid. ppp.*)



$$P(\text{gray}|+) = \frac{P(+|\text{gray}) \times P(\text{gray})}{P(+)}$$

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$



Hammond's



Gray



Dusky



Least

I'm kidding, we're not making up that many numbers

There are three key points here:

1. The math works.

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2. This is how 'we' naturally think about things! We routinely (i.e., all the freaking time) use prior information!

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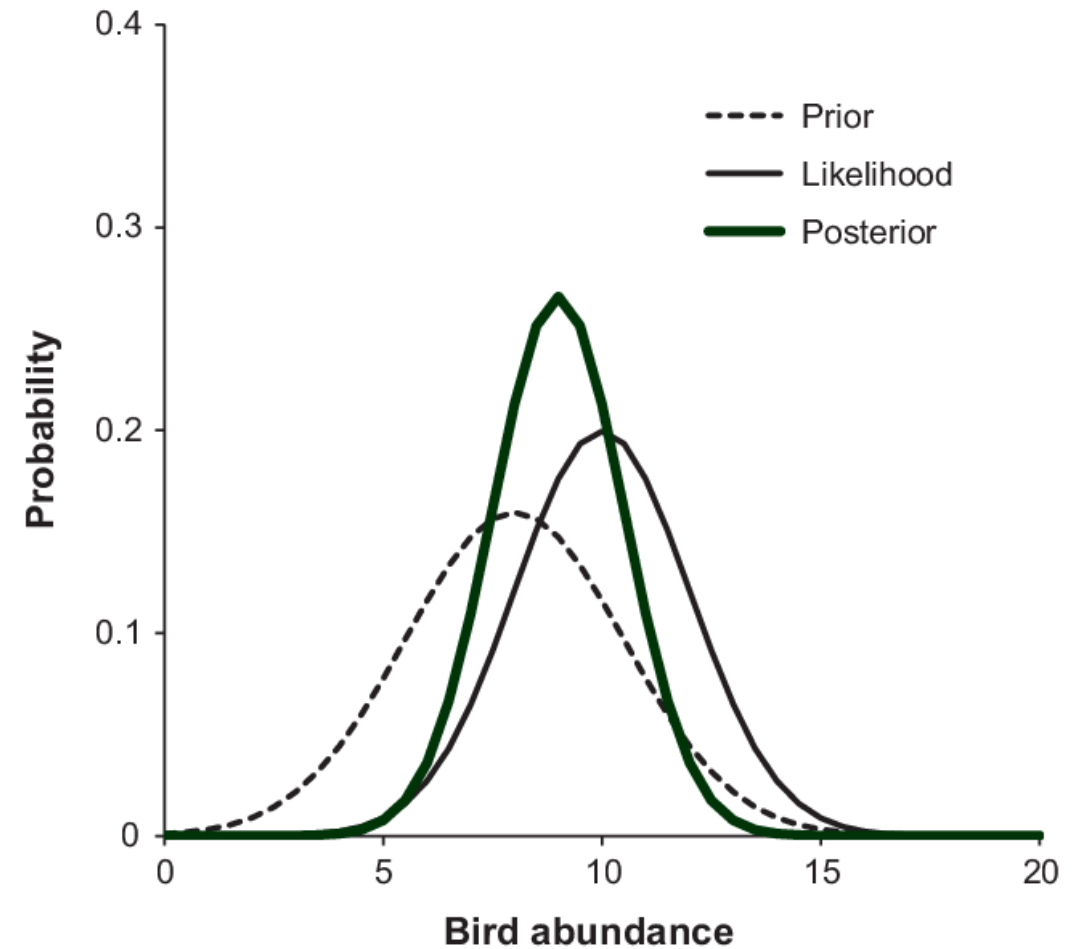
2. This is how ‘we’ naturally think about things! We routinely use prior information!

- Do we want to use a method that mimics our thinking?

3. We’re using fixed (known) values. This starts to get really complicated with uncertainty (we’re going to need help from JAGS and Stan)!

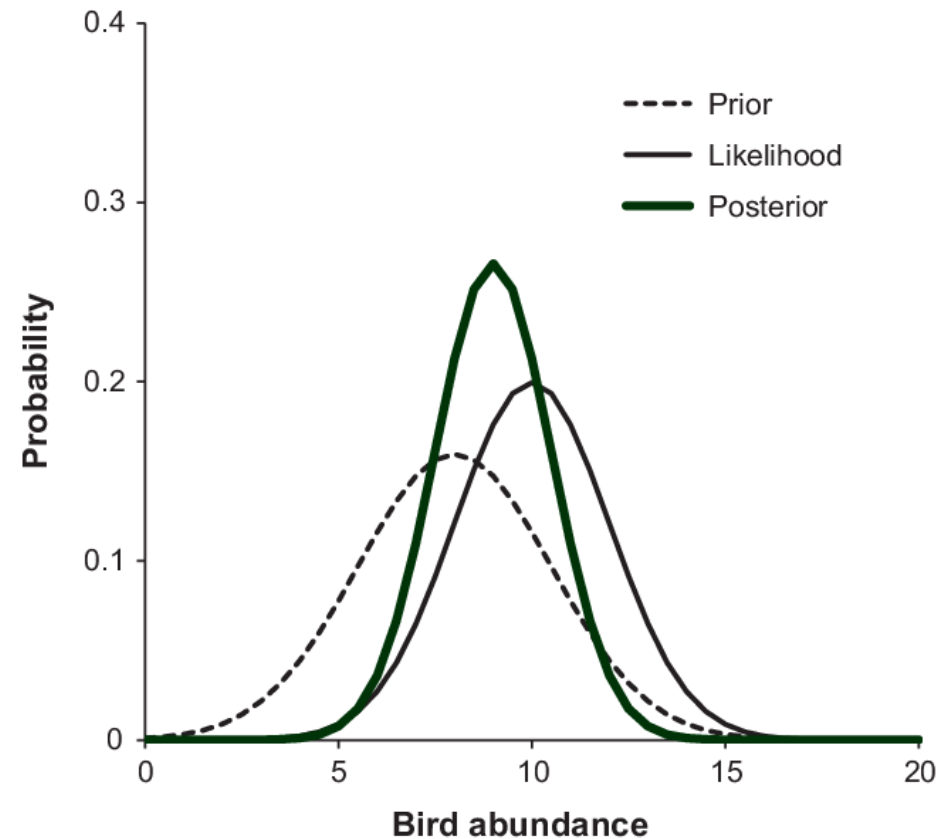
A huge non-Bayesian fear is that priors affect posteriors (inference)

- The distribution of a parameter given your prior belief and the data.



How humans learn might not be the way to go 😊

A huge non-Bayesian fear is that priors affect posteriors (inference)



Priors do affect posteriors, this is not an irrational fear

BAYES THEOREM: a response to a (the?) key criticism

We'll use 'uninformative' priors for everything. These priors will not affect our inference.

Then you can't criticize us!

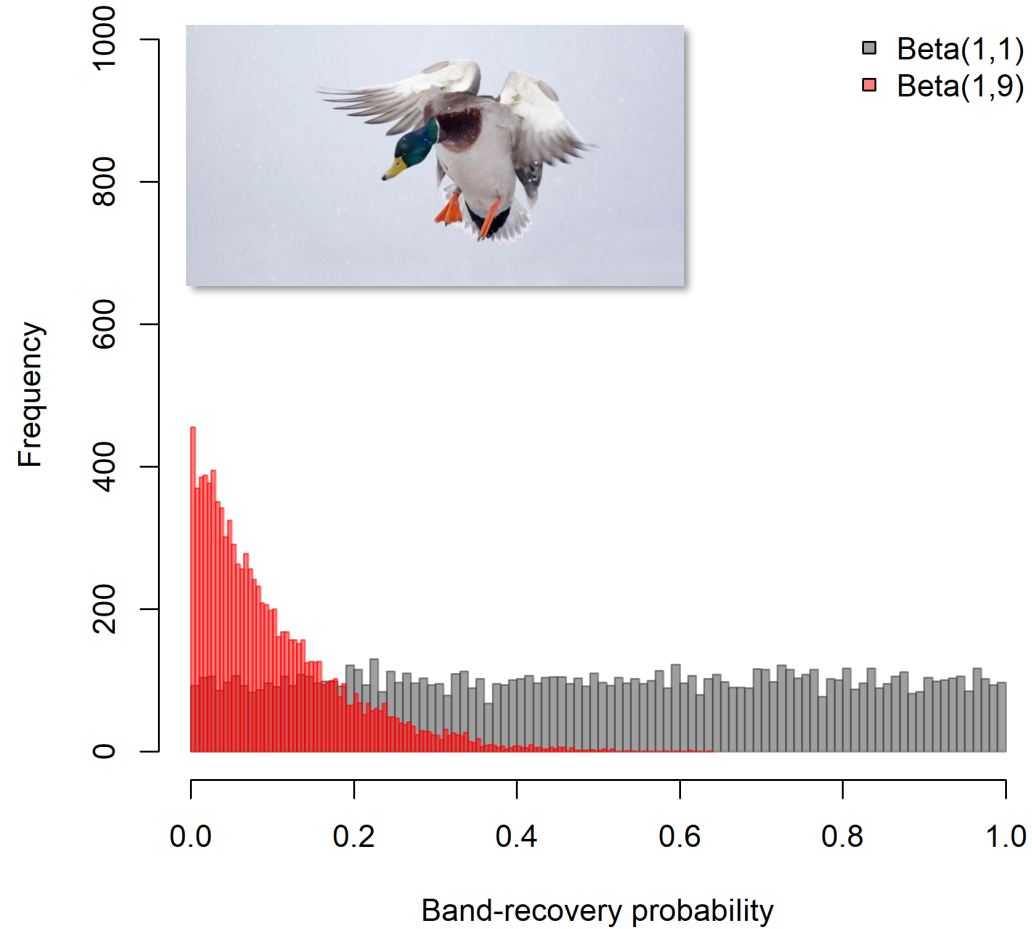
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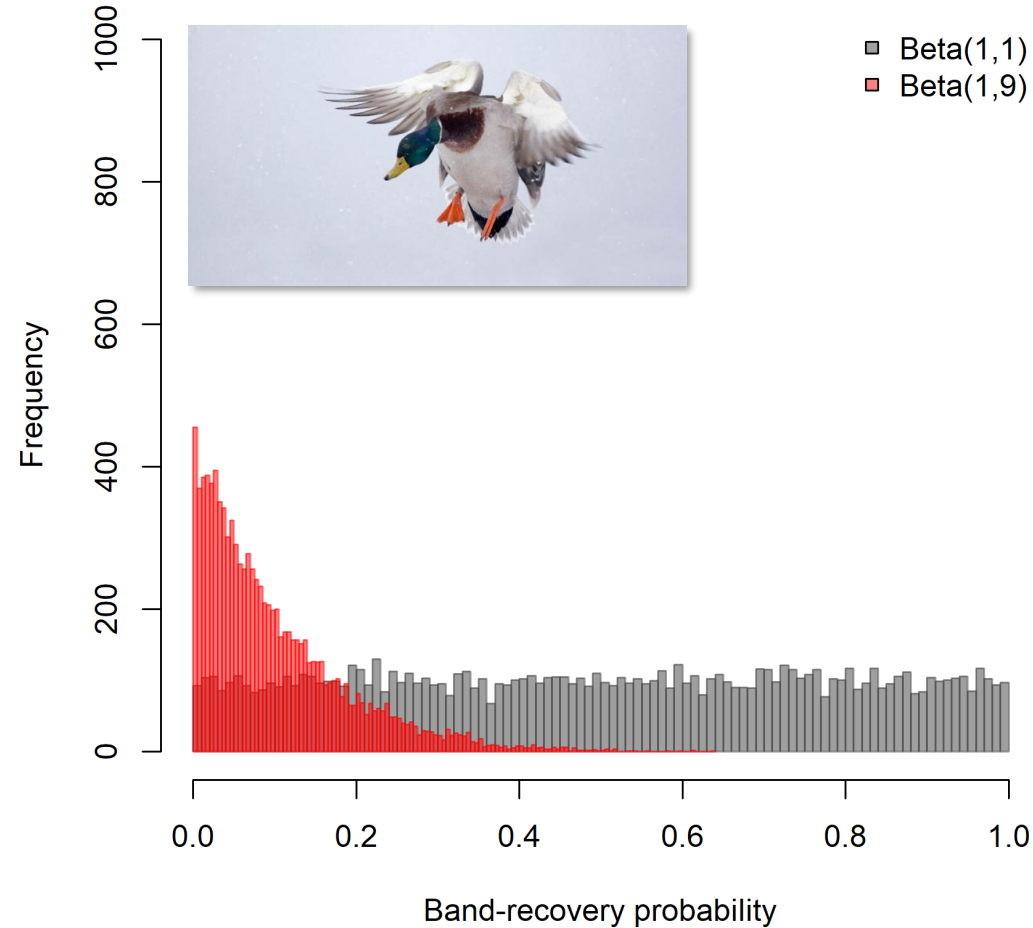
This was a key mistake, and honestly it's kind of bizarre

‘Uninformative*’ priors



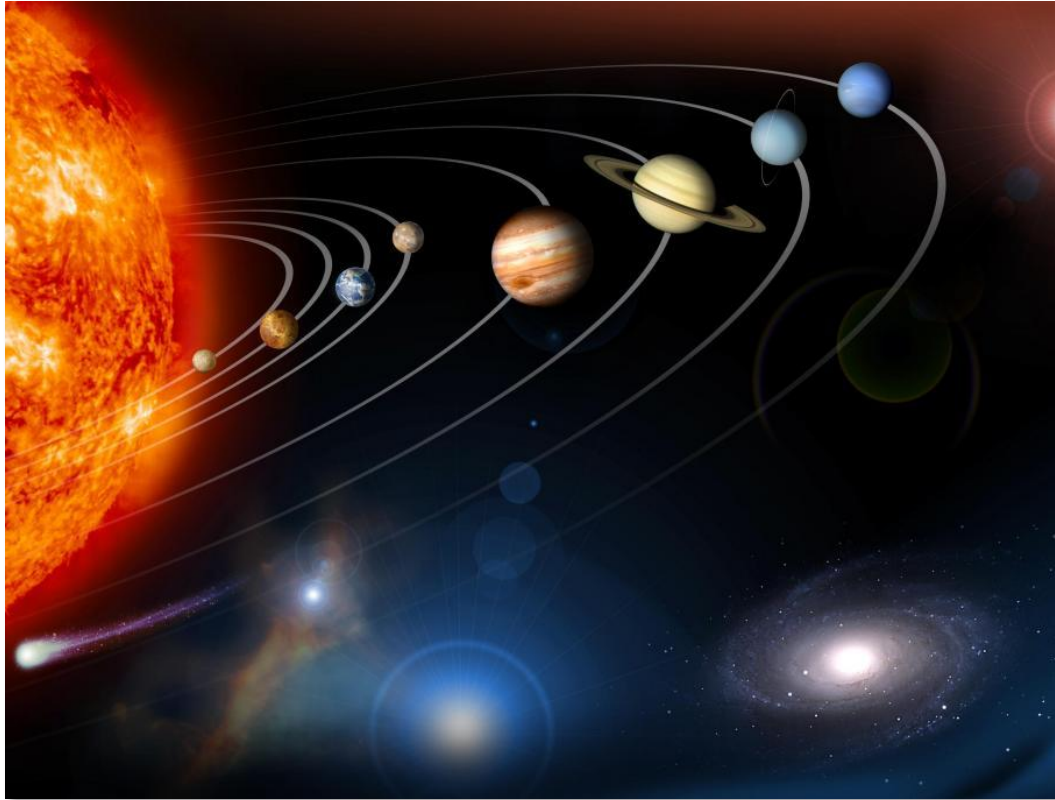
***Priors are never ‘uninformative.’**

If you disagree...

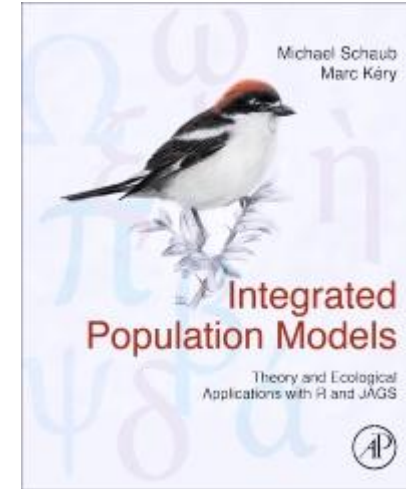
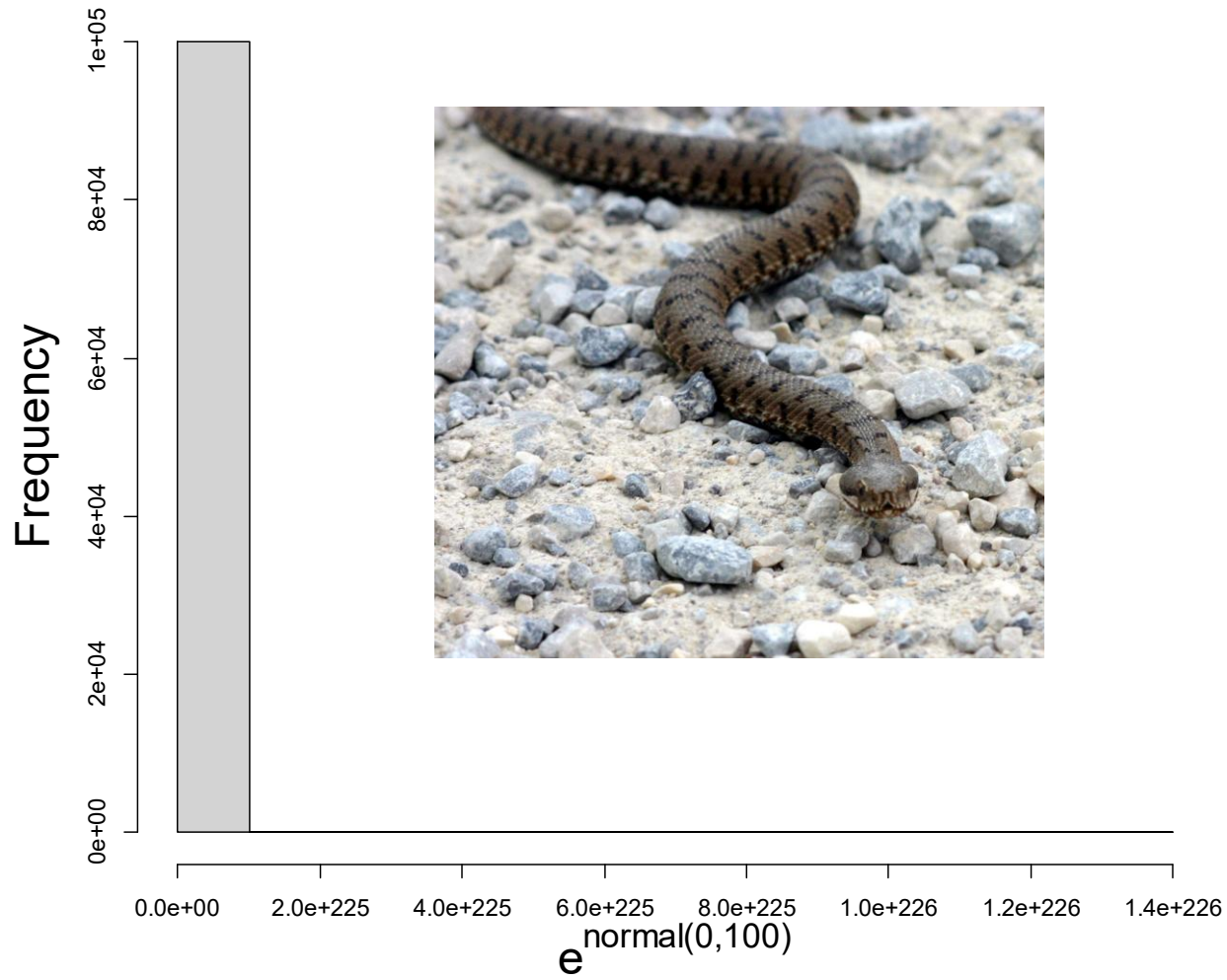


Then describe a prior without providing any information.

Sometimes priors are way too 'uninformed'



Sometimes priors are waaaaay too 'uninformed'

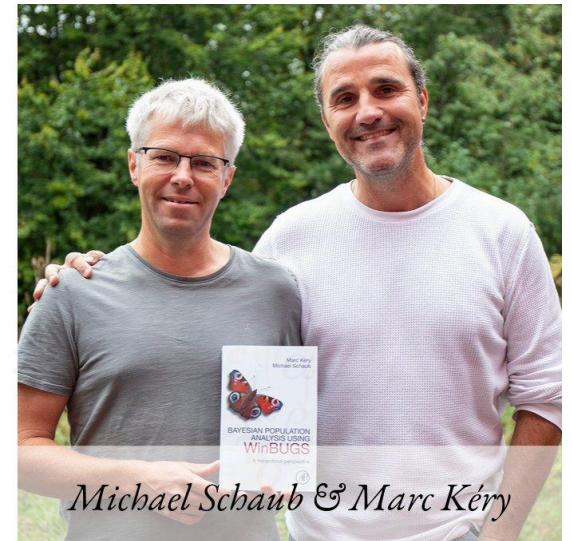


The Marsh Award for **INNOVATIVE ORNITHOLOGY**

Nominated

Michael and Marc were nominated for their groundbreaking work on Bayesian hierarchical models, changing the way we use statistics to analyse large, citizen-science data sets. The methods have not only helped BTO, but are being used worldwide on a variety of data sets and applications. Their books, as well as their workshops and teaching, only adds value to their work.

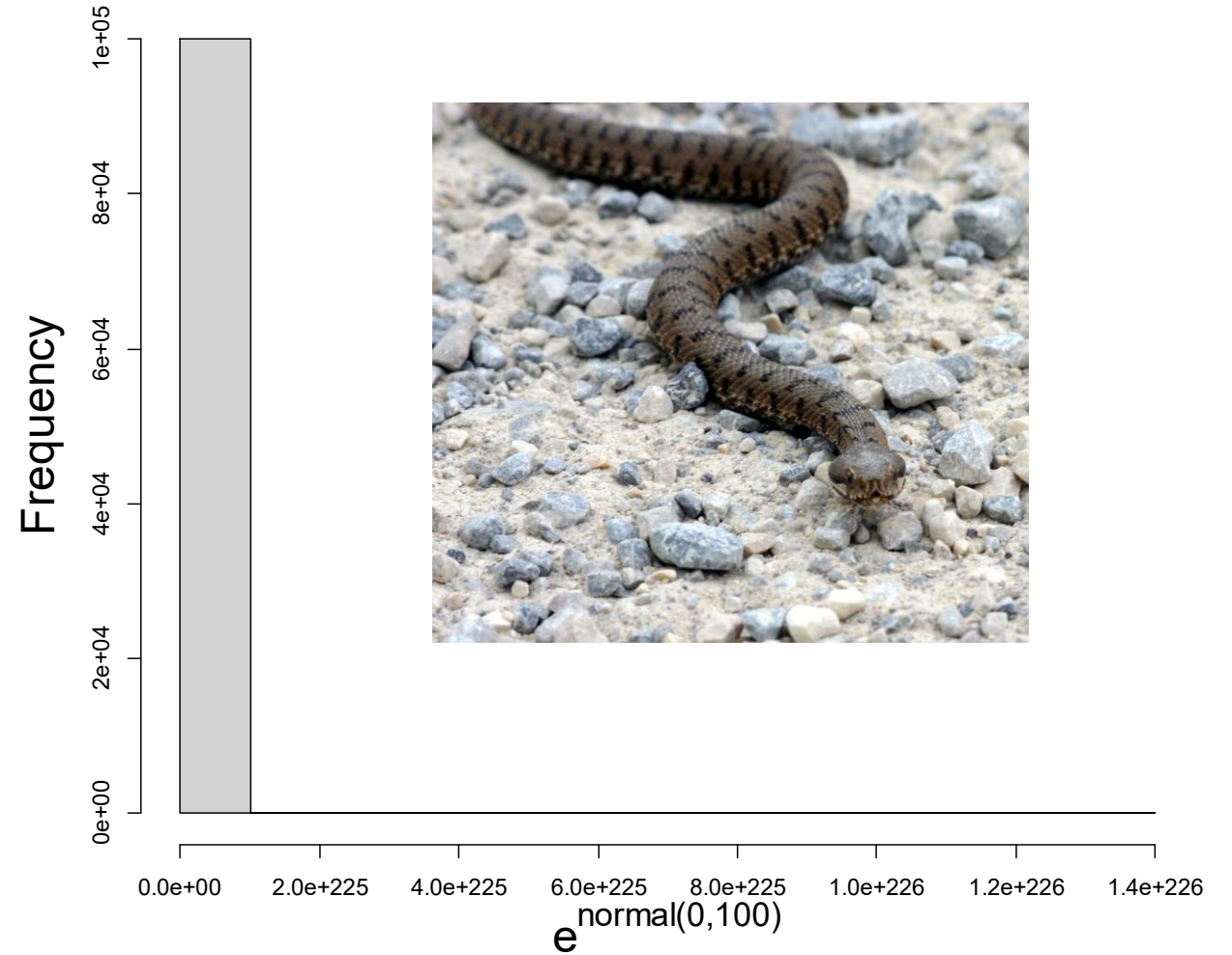
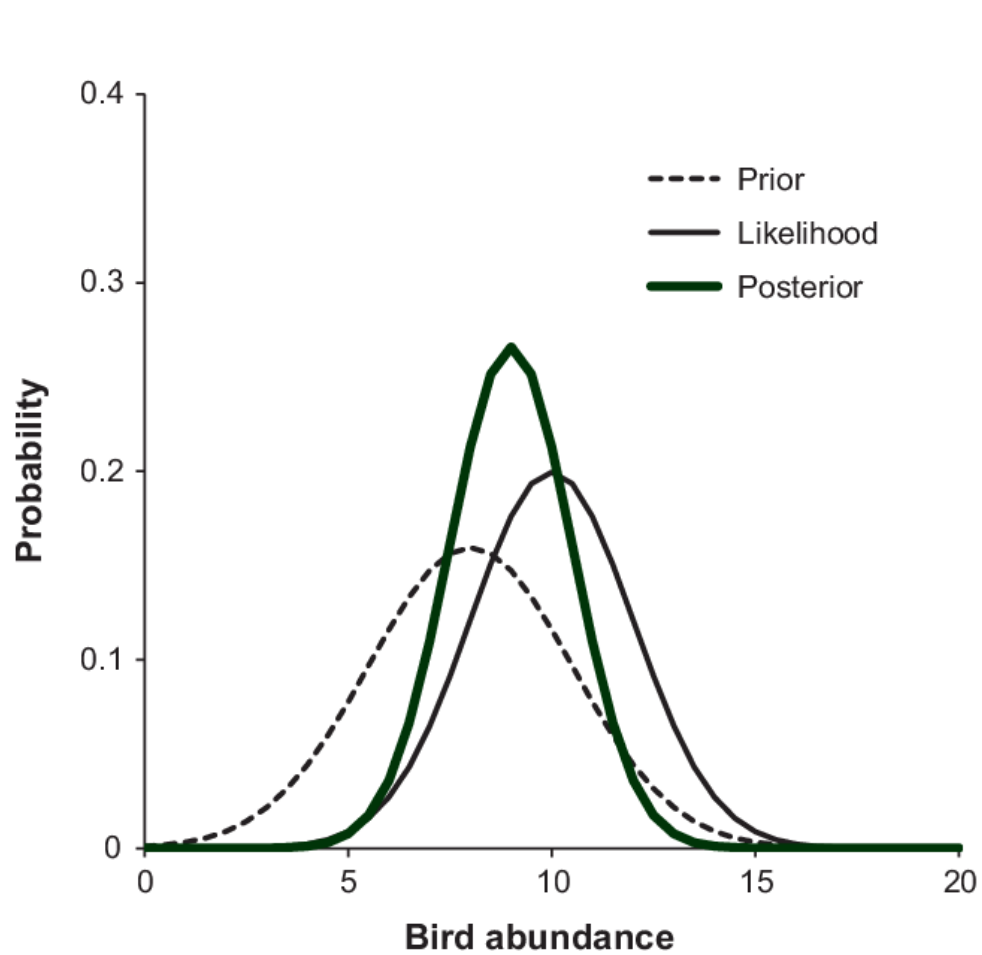
MARSH
Christian Trust



Michael Schaub & Marc Kéry

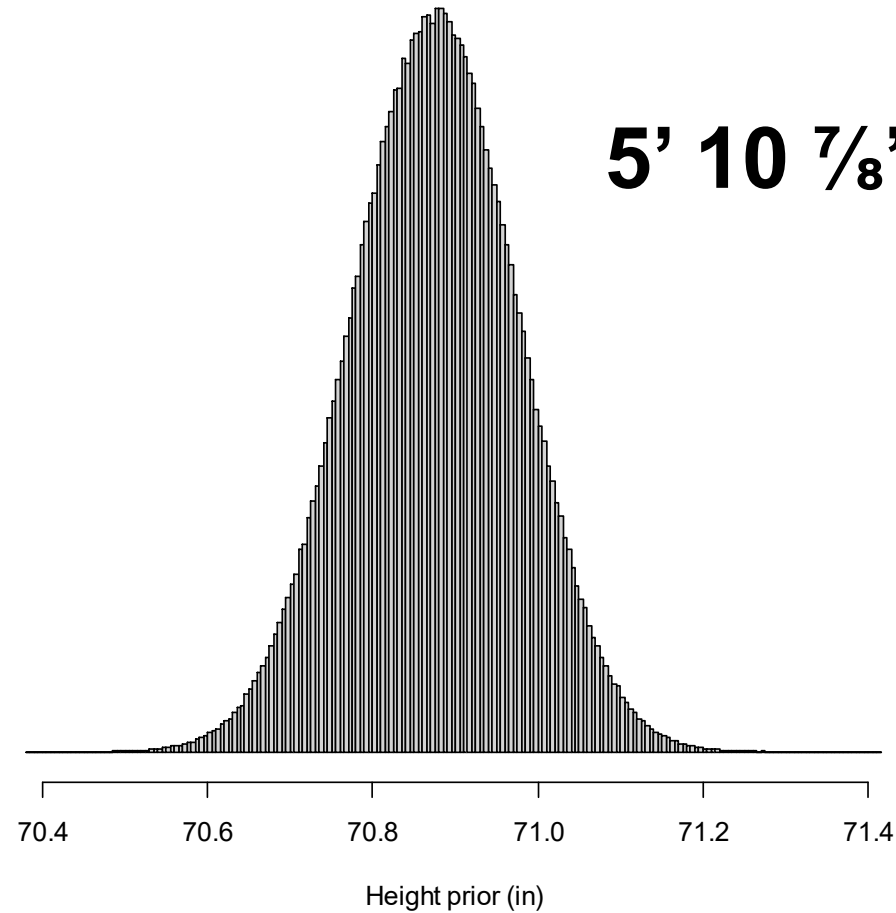
Earth's diameter is 12,756 km

Starting simple: what should our goal be with a prior?



How tall is a man you've never met before?

An example: how tall is a man you've never met before?



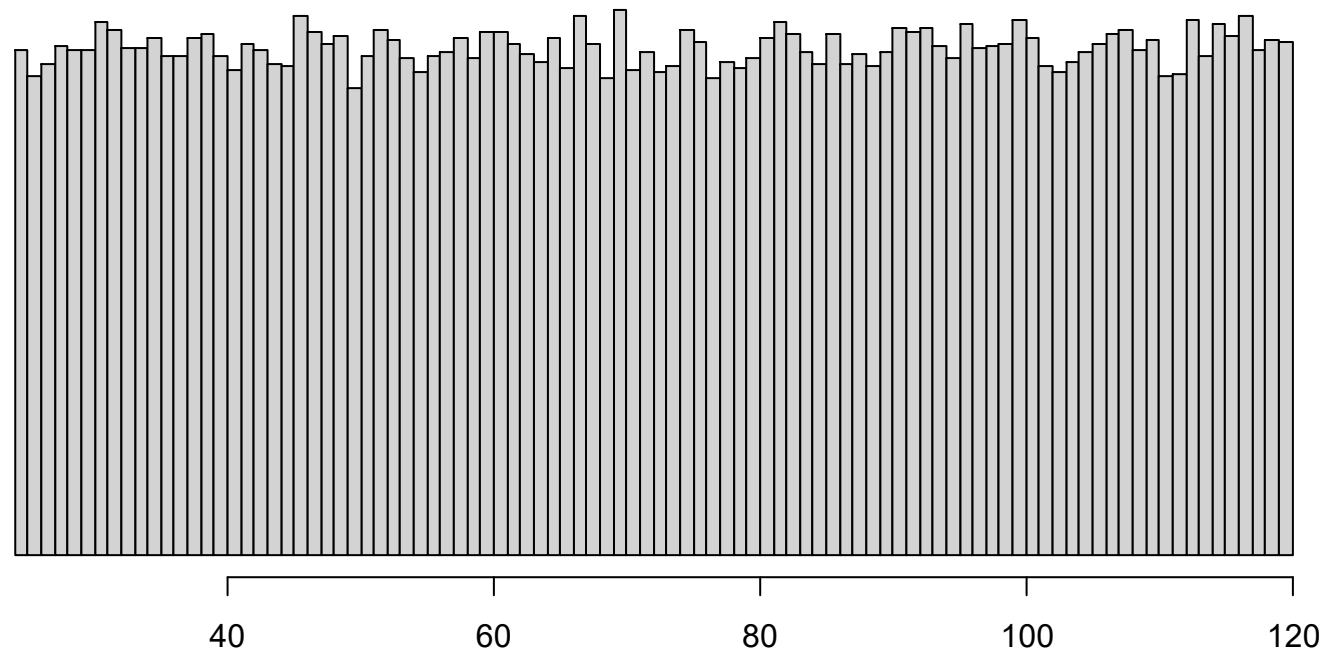
5' 10 $\frac{7}{8}$ " $\pm$ $\frac{1}{3}$ "

‘Very informative’

Normal($\mu = 70.875$, $\sigma = 0.1$)

An example: how tall is a man you've never met before?

Equally likely that they're between 2' and 10' tall



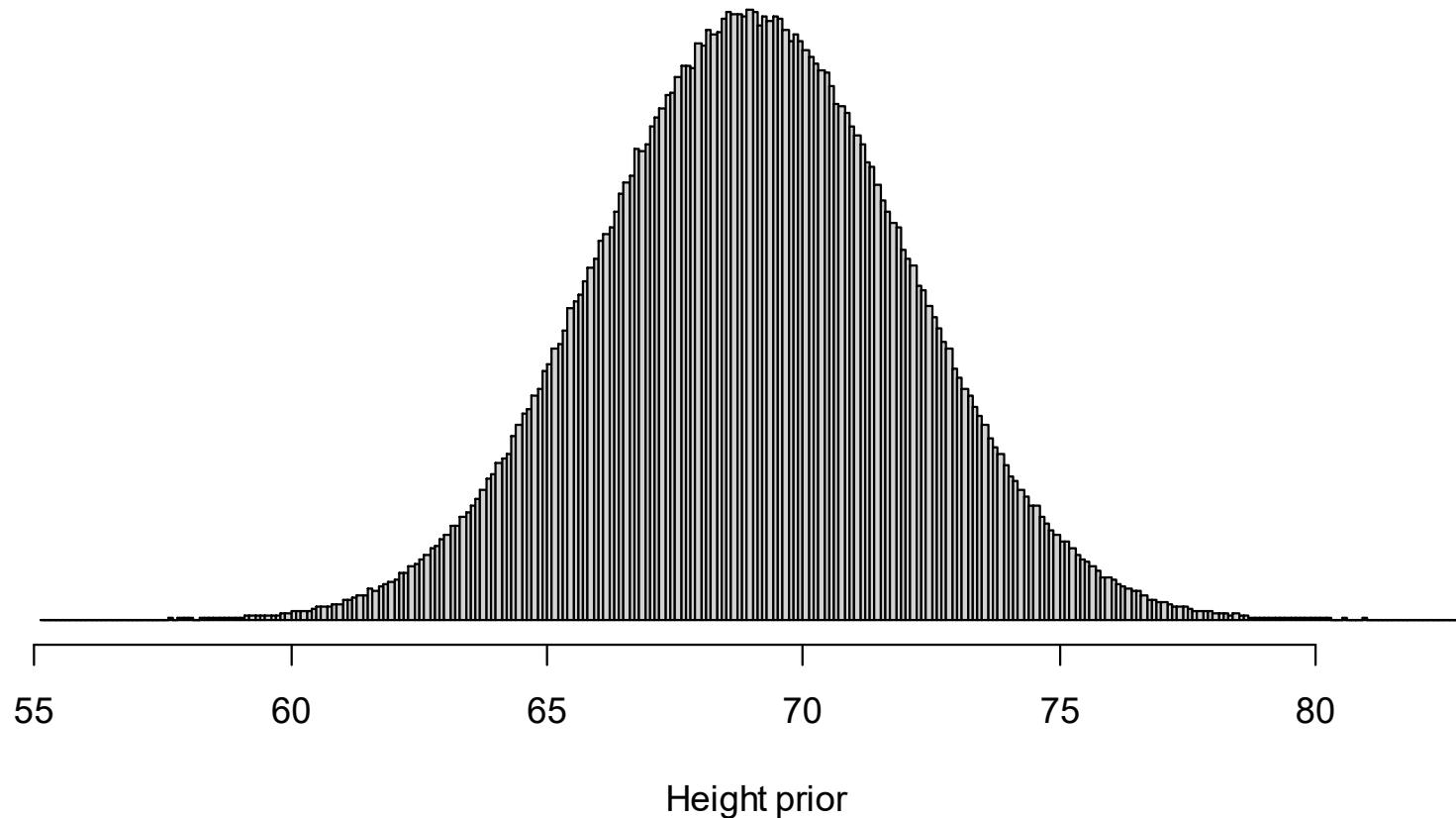
‘Uninformative’

Height prior

Uniform(24, 120)

An example: how tall is a man you've never met before?

5' 10" $\pm$ 12"



‘Biologically reasonable!’

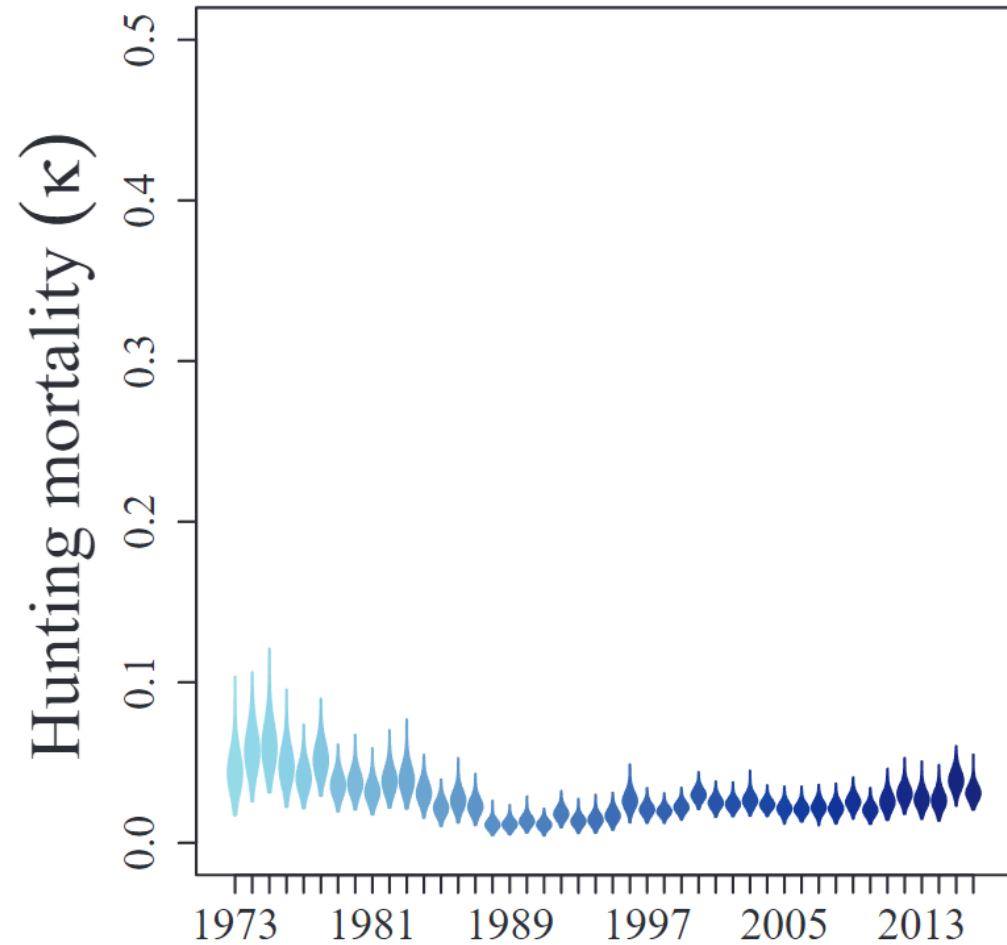
Normal($\mu = 70$, $\sigma = 3$)

An example: how tall is a man you've never met before?

‘Please stop talking to me.’

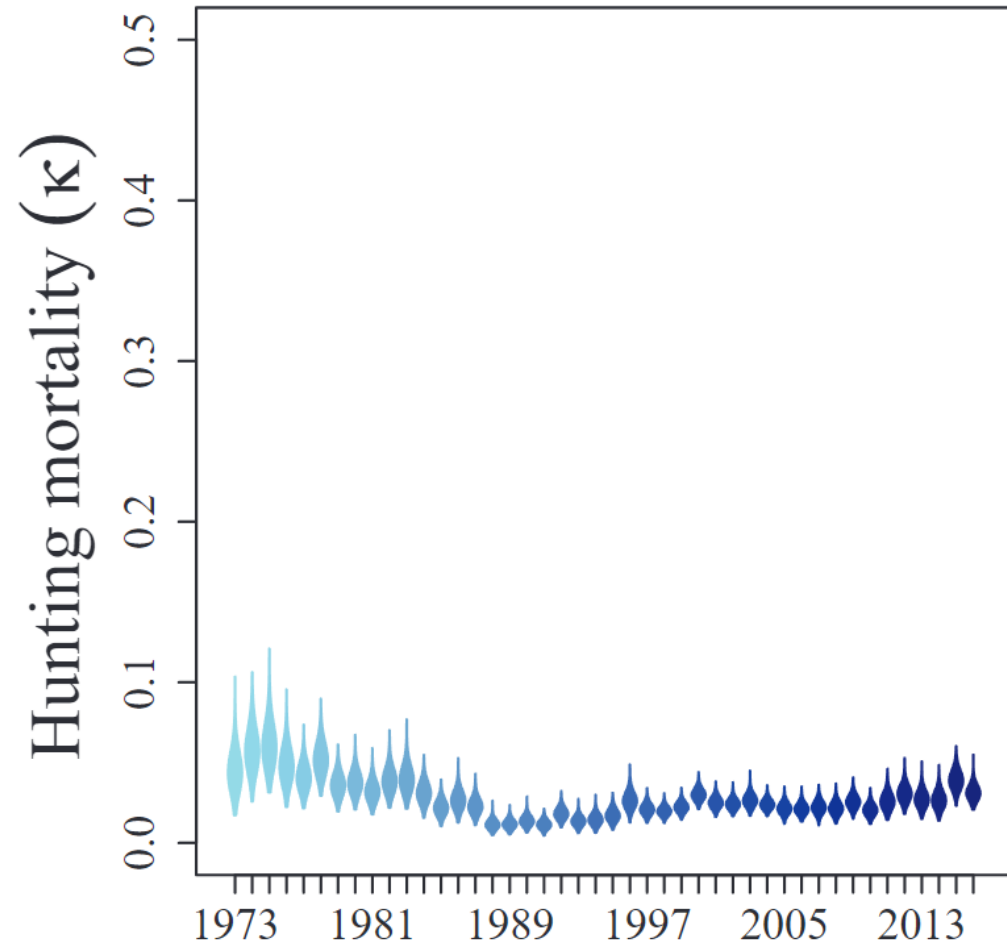
Also an acceptable answer.

Our goal: 'Biologically informative' or 'reasonably vague' priors



What is the band-recovery probability for adult female blue-winged teal?

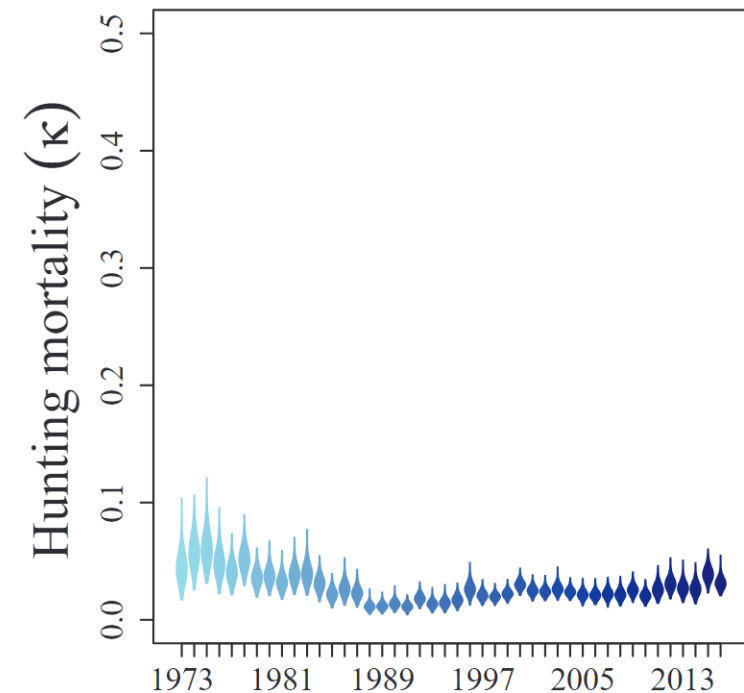
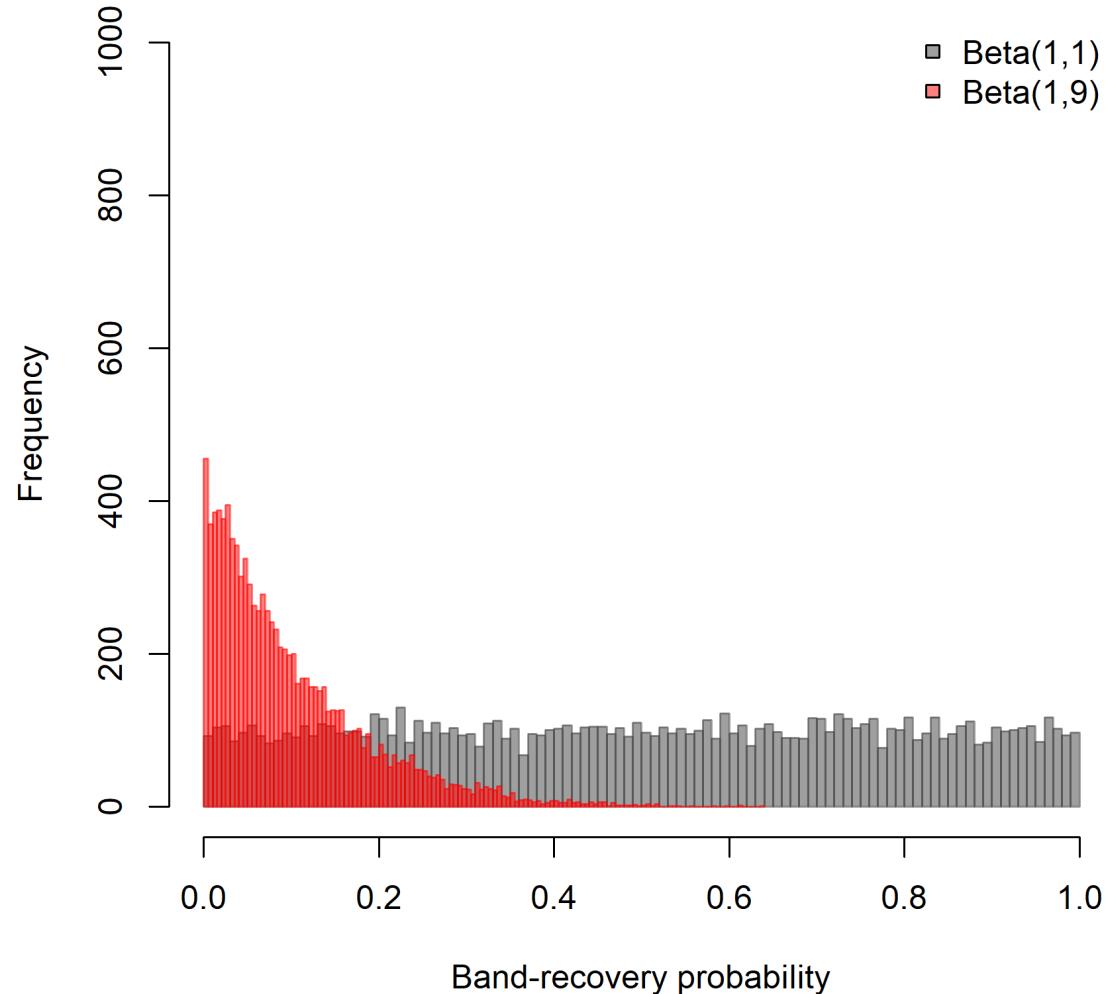
Our goal: 'Biologically informative' or 'reasonably vague' priors



What if you knew a change to hunting regulations was coming that might increase harvest above historic levels?

Our goal: 'Biologically informative' or 'reasonably vague' priors

What is the band-recovery probability for adult female blue-winged teal?



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***I recognize that this is incredibly subjective!**

Let's actually do something!

Beta-binomial models: our first Bayesian analysis

$$y \sim \text{binomial}(n, f)$$

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n: number of marked and released adult male mallards

The binomial distribution

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y: the number of recoveries of banded adult male mallards

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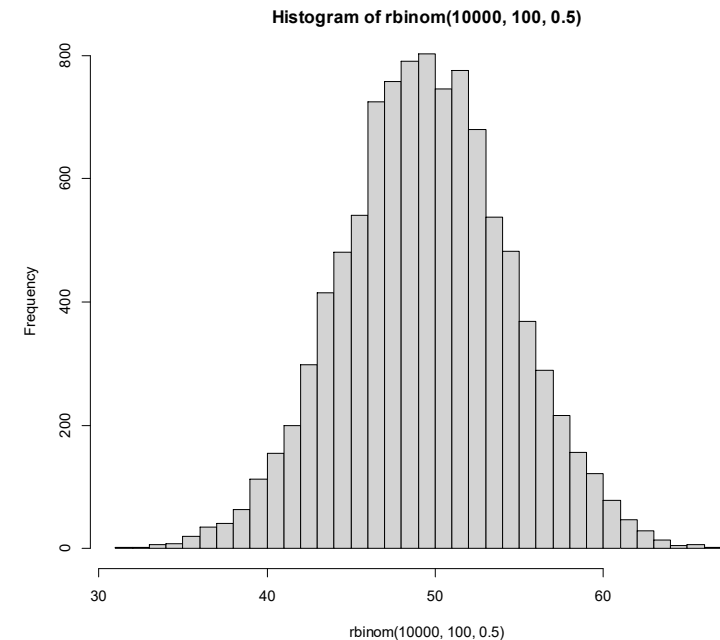
f: band-recovery probability (the parameter we want to estimate!)

The binomial distribution (and Bernoulli trials)



$$y \sim \text{binomial}(100, 0.5)$$

$$y \sim \text{Bernoulli}(0.5)$$



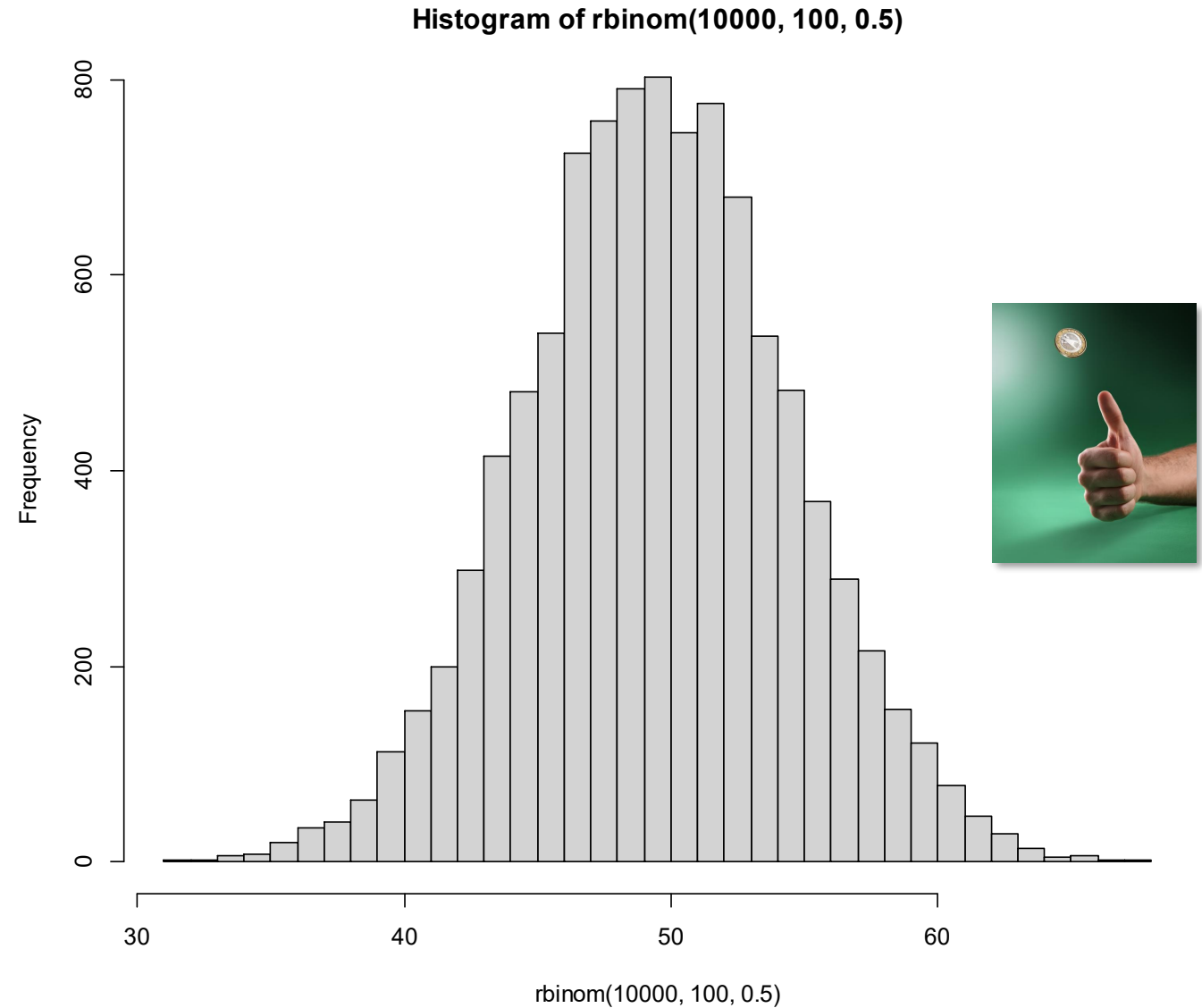
The binomial distribution (flipping coins)

$$y \sim \text{binomial}(n, p)$$

$$y \sim \text{binomial}(100, 0.5)$$

$$E(y) \sim n \times p$$

$$V(y) \sim n \times p \times (1 - p)$$



The binomial distribution

$$y \sim \text{binomial}(n, p)$$

$$y \sim \text{binomial}(40, 0.5)$$

$$E(y) \sim n \times p$$

$$V(y) \sim n \times p \times (1 - p)$$



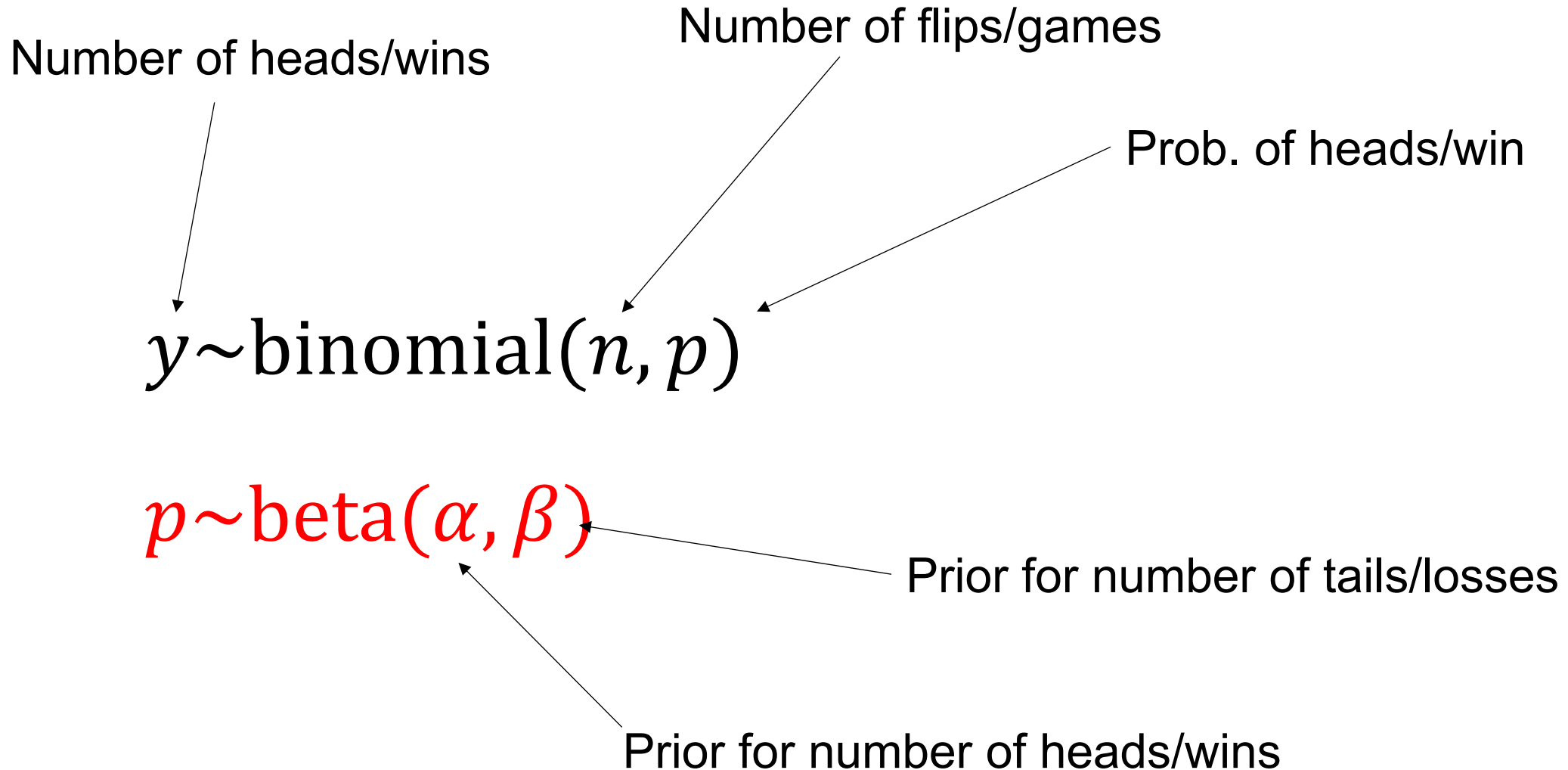
The beta distribution

$$y \sim \text{binomial}(n, f)$$

$$f \sim \text{beta}(\alpha, \beta)$$



The beta distribution as a conjugate prior for the binomial dist.



Conjugacy!*

$$y \sim \text{binomial}(n, f)$$

$$f \sim \text{beta}(\alpha, \beta)$$

Example [\[edit \]](#)

The form of the conjugate prior can generally be determined by inspection of the [probability density](#) or [probability mass function](#) of a distribution. For example, consider a [random variable](#) which consists of the number of successes s in n [Bernoulli trials](#) with *unknown* probability of success q in $[0, 1]$. This random variable will follow the [binomial distribution](#), with a probability mass function of the form

$$p(s) = \binom{n}{s} q^s (1 - q)^{n-s}$$

The usual conjugate prior is the [beta distribution](#) with parameters (α, β) :

$$p(q) = \frac{q^{\alpha-1} (1 - q)^{\beta-1}}{B(\alpha, \beta)}$$

where α and β are chosen to reflect any existing belief or information ($\alpha = 1$ and $\beta = 1$ would give a [uniform distribution](#)) and $B(\alpha, \beta)$ is the [Beta function](#) acting as a [normalising constant](#).

If we sample this random variable and get s successes and $f = n - s$ failures, then we have

$$\begin{aligned} P(s, f \mid q = x) &= \binom{s+f}{s} x^s (1 - x)^f, \\ P(q = x) &= \frac{x^{\alpha-1} (1 - x)^{\beta-1}}{B(\alpha, \beta)}, \\ P(q = x \mid s, f) &= \frac{P(s, f \mid x) P(x)}{\int P(s, f \mid y) P(y) dy} \\ &= \frac{\binom{s+f}{s} x^{s+\alpha-1} (1 - x)^{f+\beta-1} / B(\alpha, \beta)}{\int_{y=0}^1 \left(\binom{s+f}{s} y^{s+\alpha-1} (1 - y)^{f+\beta-1} / B(\alpha, \beta) \right) dy} \\ &= \frac{x^{s+\alpha-1} (1 - x)^{f+\beta-1}}{B(s + \alpha, f + \beta)}, \end{aligned}$$

which is another Beta distribution with parameters $(\alpha + s, \beta + f)$. This posterior distribution could

We can do this by hand!!

$$y \sim \text{binomial}(n, f) \quad \leftarrow \text{Data model}$$

$$f \sim \text{beta}(\alpha, \beta) \quad \leftarrow \text{Prior}$$

$$\hat{f} \sim \text{beta}(\alpha + y, \beta + (n - y)) \quad \leftarrow \text{Posterior}$$

A quick example

$$y \sim \text{binomial}(n, f) \quad \leftarrow \text{Data model}$$

$$f \sim \text{beta}(\alpha, \beta) \quad \leftarrow \text{Prior}$$

$$\hat{f} \sim \text{beta}(\alpha + y, \beta + (n - y)) \quad \leftarrow \text{Posterior}$$

A quick example

$$y \sim \text{binomial}(n, f)$$

$$137 \sim \text{binomial}(1000, f) \leftarrow \text{Data}$$

$$f \sim \text{beta}(1, 1) \leftarrow \text{Prior}$$

$$\hat{f} \sim \text{beta}(1 + 137, 1 + (863)) \leftarrow \text{Posterior}$$

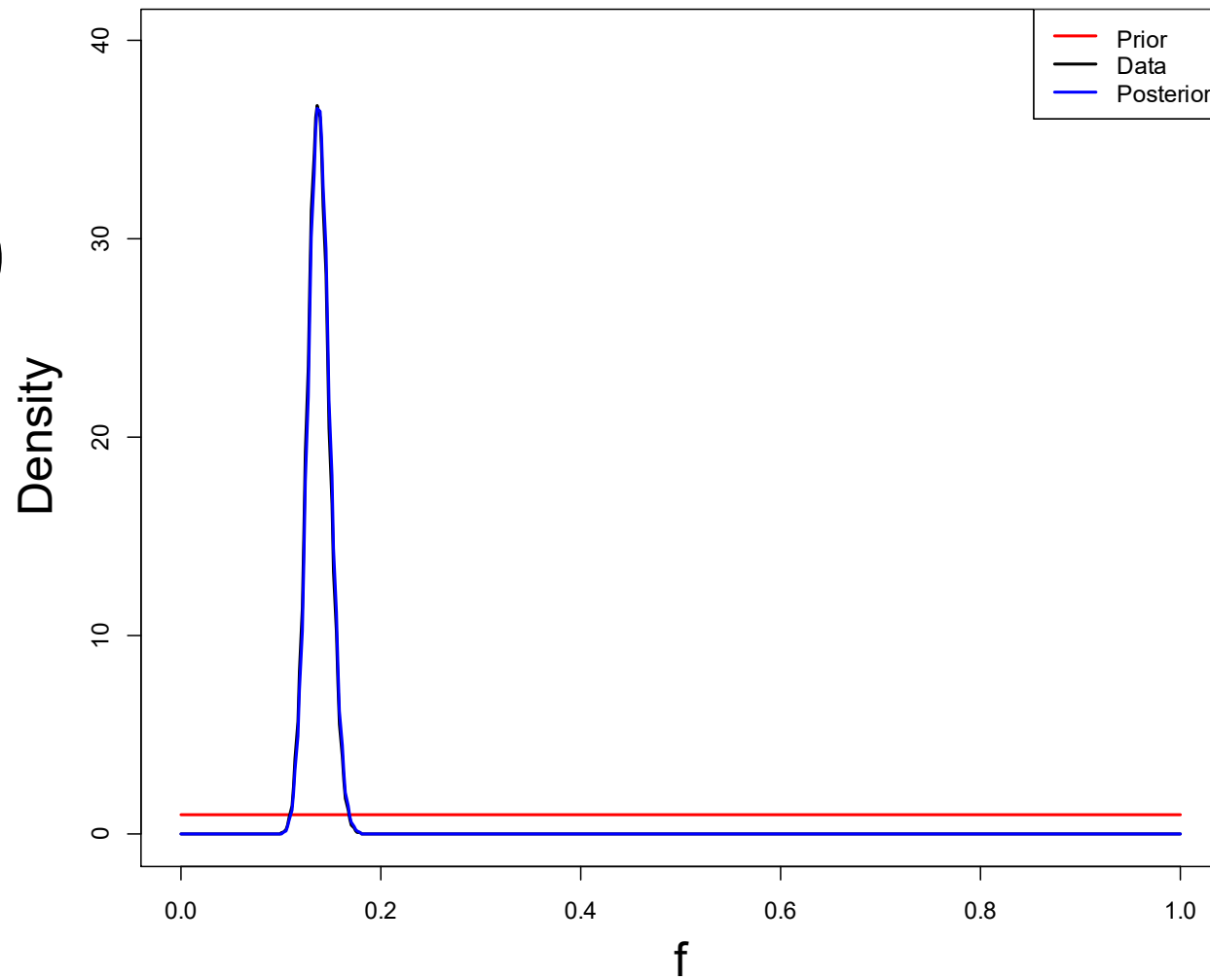
$$\hat{f} \sim \text{beta}(\alpha + y, \beta + (n - y))$$

A quick example

$137 \sim \text{binomial}(1000, f)$

$f \sim \text{beta}(1, 1)$

$\hat{f} \sim \text{beta}(138, 864)$



Playing with fire...

Weak (uniform b/w 0 and 1)

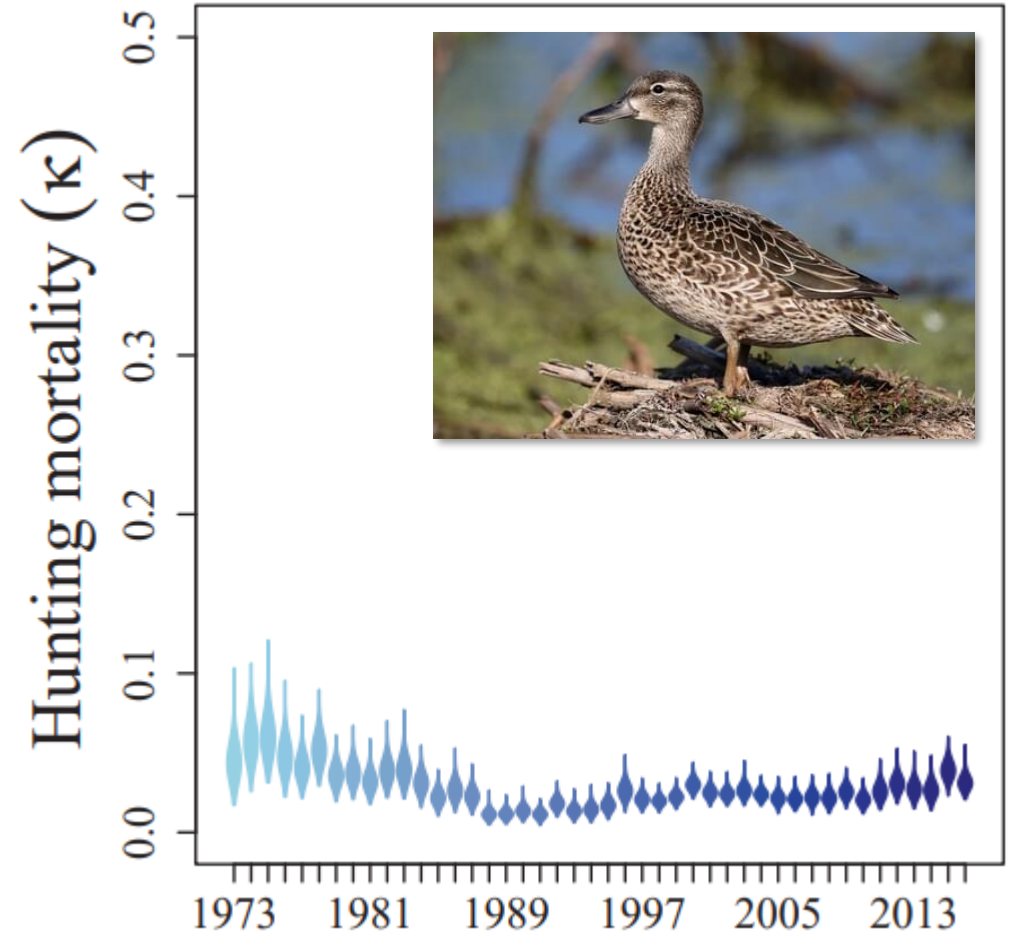
$$f \sim \text{beta}(1, 1)$$

Moderate

$$f \sim \text{beta}(1, 9)$$

Strong

$$f \sim \text{beta}(5, 95)$$



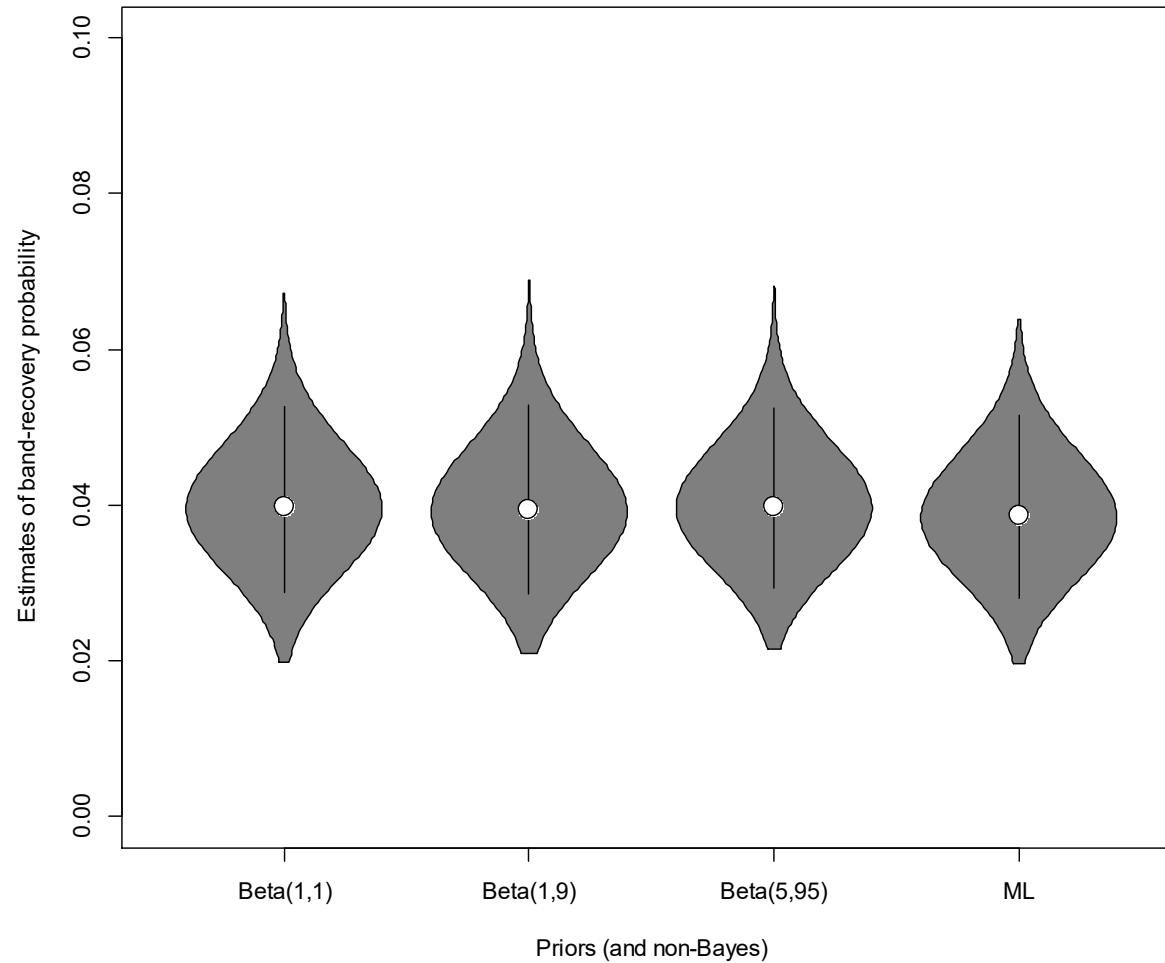
Let's try this again with some more substantial prior information

We just generated posterior distributions!

- **They are vectors of length 10000 (samples)**
- **They contain means, medians, and CREDIBLE intervals**
- **We can extract those with simple R functions (Lines 80-85)**

So what happened (Lines 88-97)?

We have strong data (and excellent prior knowledge)!



We marked 1000 ducks! The means vary by hundredths of 1 percent

What if we REALLY had strong prior knowledge

Let's assumed we're convinced we'll shoot 50 out of 1000 teal

Try $\text{beta}(50,950)$ as a 'strong' prior

What if our prior knowledge was bad?

Let's assume we're convinced that 7%, or 7 out of 100, of teal get shot!

Try $\text{beta}(7,93)$ as a 'strong' prior

What if our prior knowledge was bad?

Let's assume we're convinced that 7%, or 7 out of 100, of teal get shot!

Try $\text{beta}(70,930)$ as a 'strong' prior

Moment-matching?!

**Imagine that we have an estimate of average teal band-recovery probability.
It's reported in a manuscript as 0.0487 (sd = 0.01).**

How would we use that in a beta prior?

Moment-matching?!

If we have an estimate with $\mu=0.0487$ and $\sigma=0.01$

$$\alpha = \left(\frac{1 - \mu}{\sigma^2} - \frac{1}{\mu} \right) \mu^2$$

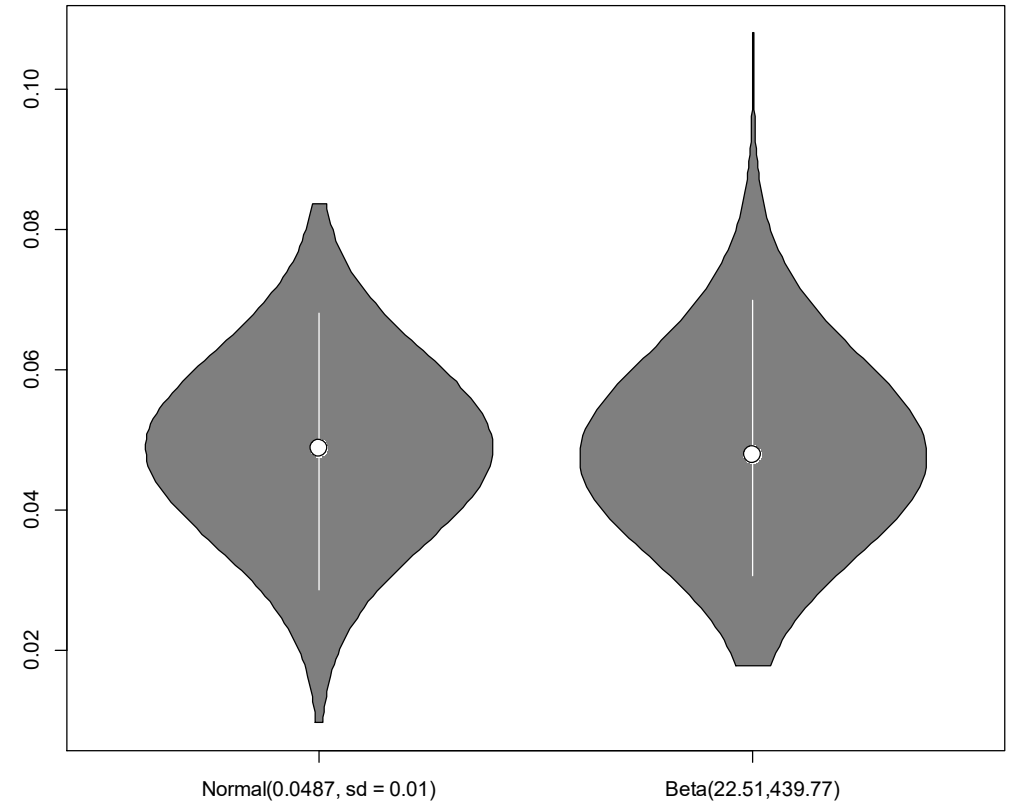
$$\beta = \alpha \left(\frac{1}{\mu} - 1 \right)$$

Moment-matching?!

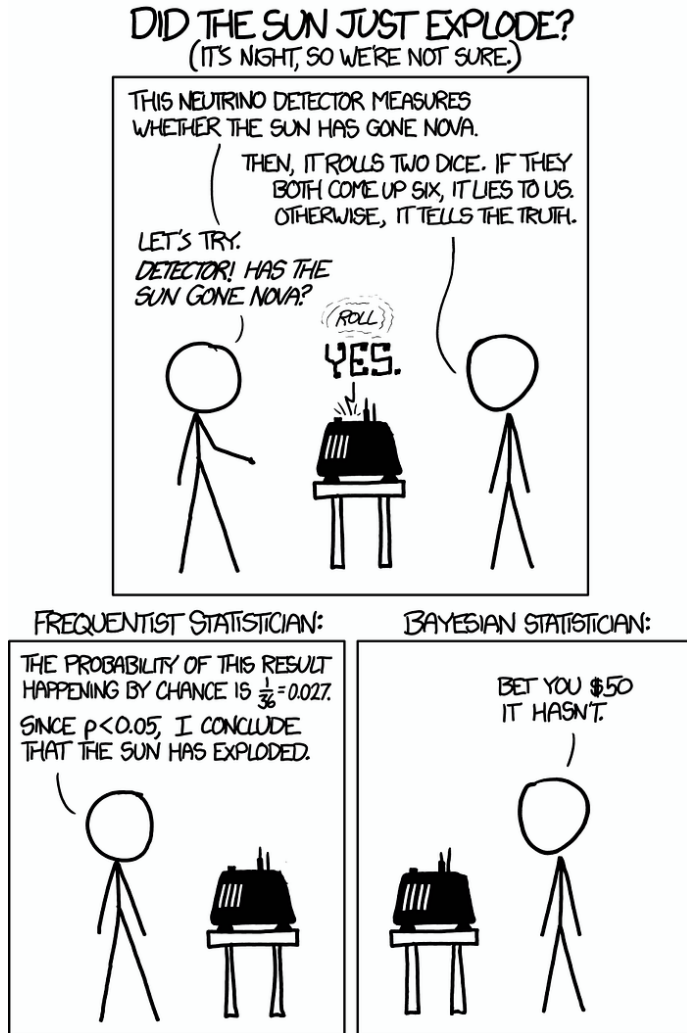
If we have an estimate with $\mu=0.0487$
and $\sigma=0.01$

$$\alpha = \left(\frac{1 - \mu}{\sigma^2} - \frac{1}{\mu} \right) \mu^2$$

$$\beta = \alpha \left(\frac{1}{\mu} - 1 \right)$$



Wrapping up...



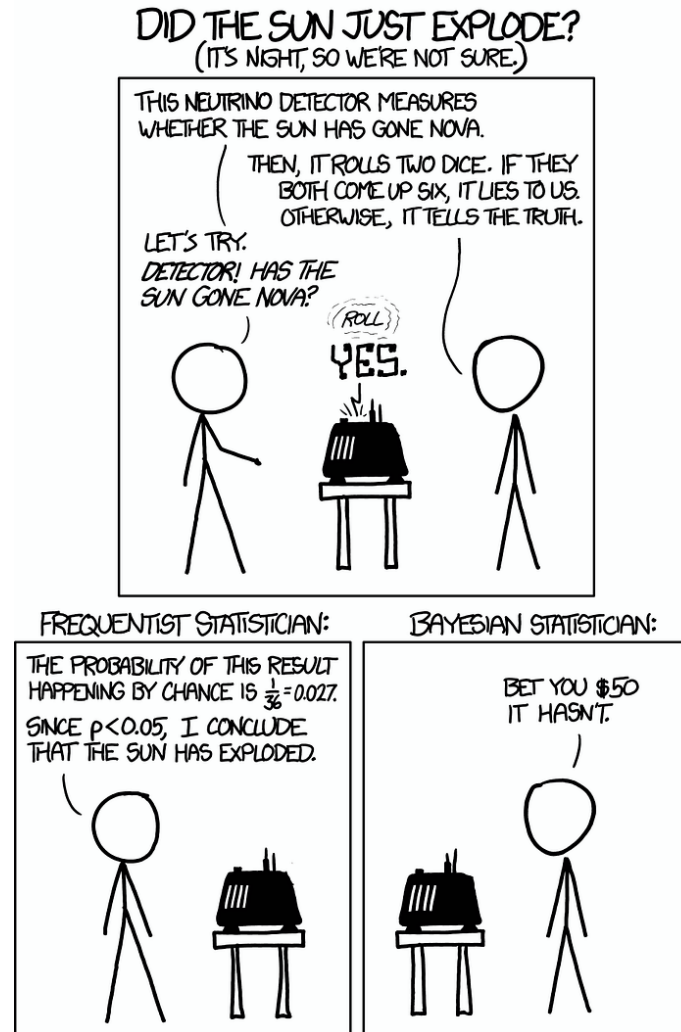
$$P(\text{sun explosion}) = 0.000000000000...000001$$

$$P(\text{sun explosion}|\text{YES}) = \frac{P(+|\text{sun explosion}) \times P(\text{sun explosion})}{P(+)}$$

Key take-home: Priors are just that, our prior belief

Key take-home: they are subjective, and cannot be uninformative

Key take-home: strong priors can save us from 'bad' data!



Key take-home: they can also hide real signals in data!

*Imagine you were convinced the right prior for our
duck harvest example was $\text{beta}(500, 500)$*

Key take-home: we can perform prior sensitivity analyses!!!

- **If our prior sensitivity analyses reveal little difference in inference between extremely vague and extremely informative priors, then our conclusions may be robust (given our model assumptions).**

Summary of summaries:

- 1. Priors are just that, our prior belief. They are ‘objectively subjective’, cannot be uninformative, and should be carefully thought about based upon existing knowledge***
- 2. Strong priors can save us from ‘bad’ data, or they can hide real signals in data!**
- 3. Prior sensitivity analysis can allow us to determine how much the strength of our beliefs impacts inference**

***sorry to be a grumpy old man here! This is important!**

Reading

<https://github.com/thomasriecke/ST595>